

# Foundations of Risk Management 目录

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# Quantitative Analysis 目录

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# Financial Markets and Products 目录

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1. Consider an exposure with a value of \$30.0 million. In L(+)/P(-) format, its expected loss is 4.0% with volatility of 15.0%; i.e.,  $\mu(L/P) = +4.0\%$ ,  $\sigma = 15.0\%$ . Conveniently, the exposure happens to have a normal distribution. Which is **nearest** to the single-period 97.5% expected shortfall (ES)?

考虑一笔名义金额为 \$30.0 百万美元的风险敞口。在 L(+)/P(-) 格式下，其预期损失为 4.0%，波动率为 15.0%；即  $\mu(L/P) = +4.0\%$ ,  $\sigma = 15.0\%$ 。为方便起见，假定该敞口的分布为正态分布。单期 97.5% 期望损失（ES）中，哪一个数值与其**最接近**？

- A. 6.60 million  
六百六十万
- B. 9.38 million  
九百三十八万
- C. 11.72 million  
一千一百七十二万
- D. 14.50 million  
一千四百五十万

答案:C

解析:

**C 选项正确。** If the distribution is normal, the 97.5% ES is approximated by the 99.0% VaR. In this case, we can use  $\$30.0 * (4.0\% + 15\% * 2.33) = \$30.0 * 38.950\% = \$11.69$ . The exact 97.5% ES is given by \$11.7210, so that is given as the correct choice.

**C 选项正确。** 若分布为正态分布，则 97.5% ES 可由 99.0% VaR 近似。此时，我们可以使用  $\$30.0 * (4.0\% + 15\% * 2.33) = \$30.0 * 38.950\% = \$11.69$ 。精确的 97.5% ES 为 \$11.7210，因此该数值被作为正确选项。

考察: Market Risk     重要程度: [■■■]

科目: Quantitative Analysis

2. Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among these (the following) four positions, which is the BEST example of wrong-way risk?

在 Zapex Financial Corporation 的金融头寸中，它写了看涨期权、进入了利率互换并写了信用违约互换。在这些（以下）四个头寸中，哪一个是最典型的反向风险？

- A. Zapex has written (aka, sold) a call option to a counterparty in the same industry  
Zapex 向同一行业的对手方出售了看涨期权
- B. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery  
Zapex 是利率互换中的固定利率支付方，在经济复苏期间
- C. Zapex has written (aka, sold) a credit default swap (CDS) to counterparty whose credit quality is positively correlated with the reference entity  
Zapex 向信用质量与参考实体正相关的对手方出售了信用违约互换
- D. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability  
Zapex 在远期货币合约中持有正相关于对手方违约概率的标记市值收益

答案:D

解析:

**D 选项正确。** Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability.

Wrong-way (credit) risk is adverse correlation between the exposure and the credit quality, so wrong-way risk is a positive (negative) correlation between exposure and default probability (credit quality). This is wrong-way risk because greater exposure is associated with a lower likelihood of payment. Algebraically, wrong-way (or right-way) risk is some dependence between exposure/EAD and PD. We normally write  $EL\% = PD \times EAD$  but that assumes independence between PD and EAD, so a way to think about WWR is simply when  $EL > PD \times EAD$ , because  $EL \neq PD \times EAD$  is also a condition for non-independence.

In regard to (A), (B) and (C), each is an example of RIGHT-WAY risk.

考察: Credit Risk     重要程度: [■ ■ □]

科目: Financial Markets and Products

3. In December 2017, the Basel Committee (BCBS) made several revisions to the Basel III framework. This included revisions to both the standardized approach to credit risk (“Credit risk accounts for the bulk of most banks’ risk-taking activities and hence their regulatory capital requirements. The standardized approach is used by the majority of banks around the world, including in non-Basel Committee jurisdictions,” wrote the Committee) and the internal ratings-based (IRB) approaches to credit risk (“The financial crisis highlighted a number of shortcomings related to the use of internally modeled approaches for regulatory capital, including the IRB approaches to credit risk. These shortcomings include the excessive complexity of the IRB approaches, the lack of comparability in banks’ internally modeled IRB capital requirements and the lack of robustness in modelling certain asset classes,” wrote the Committee).<sup>1</sup>

In regard to these December 2017 revisions to credit risk approaches in Basel III, which of the following statements is TRUE?

<sup>1</sup> “High-level summary of Basel III reforms,” (Basel Committee on Banking Supervision Publication, December 2017)

2017 年 12 月，巴塞尔委员会（BCBS）对巴塞尔协议 III 框架进行了多项修订。其中包括对信用风险标准法的修订（委员会写道：“信用风险占据了大多数银行风险承担活动以及相应监管资本要求的主要部分。标准法被世界上大多数银行采用，包括那些不在巴塞尔委员会成员辖区内的银行”），以及对信用风险内部评级法（IRB）方法的修订（委员会写道：“金融危机凸显了与监管资本内部模型方法使用相关的一些缺陷，其中包括信用风险内部评级法（IRB）方法。这些缺陷包括 IRB 方法的过度复杂性、各银行内部模型化的 IRB 资本要求缺乏可比性，以及在对某些资产类别建模时缺乏稳健性”）。<sup>1</sup> 关于 2017 年 12 月对巴塞尔 III 中信用风险计量方法所作的这些修订，以下哪一项表述是正确的？<sup>1</sup> “High-level summary of Basel III reforms”（《巴塞尔协议 III 改革的高层次概要》），巴塞尔银行监管委员会出版物，2017 年 12 月

A. In the standardized approach (SA) for residential real estate exposures, the risk weights are lower if the loan-to-value (LTV) falls into a lower band

在住宅房地产敞口中，标准法（SA）下，如果贷款价值比（LTV）落入较低的区间，则风险权重较低

B. The standardized approach (SA) eliminated the treatment of unrated exposures because all exposures are required to have an external credit rating

标准法（SA）消除了对无评级敞口的治疗，因为所有敞口都必须具有外部信用评级

C. The internal ratings-based (IRB) approach removed the option to use the foundation IRB (F-IRB) because the advanced IRB (A-IRB) must be used for all asset classes

内部评级法（IRB）消除了使用基础 IRB（F-IRB）的选项，因为所有资产类别都必须使用高级 IRB（A-IRB）

D. The internal ratings-based (IRB) approach removed the parameter input floors (in the case of PD, LGD, and EAD) to increase RWA variability across different banks

内部评级法（IRB）消除了参数输入地板（在 PD、LGD 和 EAD 的情况下），以增加不同银行之间的 RWA 可变性

答案:A

解析:

**Option A is correct.** In the standardized approach (SA) for residential real estate exposures, the risk weights is lower if the loan-to-value (LTV) falls into a lower band.

See below. Writes BCBS (High-level summary of Basel III reforms, December 2017): “For residential real estate exposures, more risk-sensitive approaches have been developed, whereby risk weights vary based on the LTV ratio of the mortgage (instead of the existing single risk weight) and in ways that better reflect differences in market structures.”

| Residential real estate exposures                |           |                    |            |            |            |             |                    |                    |
|--------------------------------------------------|-----------|--------------------|------------|------------|------------|-------------|--------------------|--------------------|
| LTV bands                                        | Below 50% | 50% to 60%         | 60% to 70% | 70% to 80% | 80% to 90% | 90% to 100% | above 100%         | Criteria not met   |
| General RRE                                      |           |                    |            |            |            |             |                    |                    |
| Whole loan approach RW                           | 20%       | 25%                | 30%        | 40%        | 50%        | 70%         | RW of counterparty |                    |
| Loan-splitting approach <sup>2</sup> RW          | 20%       | RW of counterparty |            |            |            |             |                    | RW of counterparty |
| Income-producing residential real estate (IPRRE) |           |                    |            |            |            |             |                    |                    |
| Whole loan approach RW                           | 30%       | 35%                | 45%        | 60%        | 75%        | 105%        | 150%               |                    |

In regard to (B), (C) and (D), each is FALSE. Instead, the following are true:

- The Basel III reforms included a more granular approach to unrated exposures and reduced the “mechanistic reliance on credit ratings”
- In regard to the internal ratings-based approaches, GARP writes, “the financial crisis highlighted a number of shortcomings related to the use of internally modeled approaches for regulatory capital, including the IRB approaches to credit risk. These shortcomings include the excessive complexity of the IRB approaches, the lack of comparability in banks’ internally modeled IRB capital requirements and the lack of robustness in modelling certain asset classes. To address these shortcomings, the Committee has made the following revisions to the IRB approaches: (i) removed the option to use the advanced IRB (A-IRB) approach for certain asset classes; (ii) adopted “input” floors (for metrics such as probabilities of default (PD) and loss-given-default (LGD)) to ensure a minimum level of conservatism in model parameters for asset classes where the IRB approaches remain available; and (iii) provided greater specification of parameter estimation practices to reduce RWA variability.” — Source: 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition.
- In regard to the Specification of Input Floors: “The revised IRB framework also introduces minimum floor values for bank-estimated IRB parameters that are used as inputs to the calculation of RWA. These include PD floors for both the F-IRB and A-IRB approaches, and LGD and EAD floors for the A-IRB approach. In some cases, these floors consist of recalibrated values of the existing Basel II floors. In other cases, the floors represent new constraints for banks’ IRB models.” — Source: 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition.

考察: Operational Risk and Resiliency      重要程度: [■■■□]

科目: Quantitative Analysis

4. An asset that is not very liquid is quoted bid \$89.00, offer \$96.00. Which is **nearest** to the proportional bid-offer spread?  
一笔不太流动的资产的报价为买入价 \$89.00，卖出价 \$96.00。哪一个数值与其**最接近**的相对买卖价差？

- A. 0.0378
- B. 0.0757
- C. 0.1538
- D. 7.000

答案:B

解析:

B 选项正确。  $0.0757$ . Because  $(96.00 - 89.00)/92.50 = 0.075676$

According to Hull (24.1.2. Measuring Market Liquidity), the two measures of bid-offer spread are:

- Dollar bid-offer spread =  $p$  = Offer price – Bid price
- Proportional bid-offer spread =  $s$  = (Offer price – Bid price) / Mid-market price

(Source: John C. Hull, Risk Management and Financial Institutions, 5th Edition (Hoboken, NJ: John Wiley & Sons, 2018))

考察: Market Liquidity      重要程度: [■■■□]

科目: Quantitative Analysis

5. Richard conducted a factor regression for his firm’s Flagship Portfolio. He used the three-factor Fama-French specification where the factors are MKT (the market), SMB (small stocks minus big stocks), and HML (value stocks minus growth stocks). His factor regression results are displayed below.



**Flagship excess returns,  $R(p) - R_f$ ,  
Fama-French benchmark (monthly)**

|                                   | <b>Coeffi-<br/>cient</b> | <b>Standard<br/>Error (SE)</b> |
|-----------------------------------|--------------------------|--------------------------------|
| <b>Alpha, <math>\alpha</math></b> | <b>0.300%</b>            | <b>0.033%</b>                  |
| <b>MKT Loading</b>                | <b>1.250</b>             | <b>0.096</b>                   |
| <b>SMB Loading</b>                | <b>0.640</b>             | <b>0.043</b>                   |
| <b>HML Loading</b>                | <b>-0.920</b>            | <b>0.066</b>                   |
| <b>Adjusted <math>R^2</math></b>  | <b>0.290</b>             |                                |

Please note the regression is based on monthly returns. Each of the following statements is TRUE about these factor regression results EXCEPT which is false?

Richard 对其公司旗舰组合进行了因子回归。他采用了三因子 Fama-French 模型设定，其中因子包括 MKT（市场）、SMB（小盘股减大盘股）以及 HML（价值股减成长股）。其因子回归结果如下所示。

**Flagship excess returns,  $R(p) - R_f$ ,  
Fama-French benchmark (monthly)**

|                                   | <b>Coeffi-<br/>cient</b> | <b>Standard<br/>Error (SE)</b> |
|-----------------------------------|--------------------------|--------------------------------|
| <b>Alpha, <math>\alpha</math></b> | <b>0.300%</b>            | <b>0.033%</b>                  |
| <b>MKT Loading</b>                | <b>1.250</b>             | <b>0.096</b>                   |
| <b>SMB Loading</b>                | <b>0.640</b>             | <b>0.043</b>                   |
| <b>HML Loading</b>                | <b>-0.920</b>            | <b>0.066</b>                   |
| <b>Adjusted <math>R^2</math></b>  | <b>0.290</b>             |                                |

请注意，该回归是基于月度收益率进行的。以下关于这些因子回归结果的陈述中，除了哪一项为假，其余均为真

- A. The portfolio's style appears to growth-oriented, small-cap  
该组合的风格似乎是成长型、小盘股
- B. This portfolio manager(s) appear to be skillful; that is, the alpha is significant  
该组合经理似乎很擅长；即，alpha 是显著的
- C. The low adjusted  $R^2$  and low standard errors imply these factors are not appropriate to this portfolio  
低调整后的  $R^2$  和低标准误差表明这些因子不适合该组合
- D. The implied CAPM benchmark is leveraged as it borrows cash (i.e., T-bills) to invest more than 100% (i.e.,  $> 1.0$ ) in the market portfolio  
隐含的 CAPM 基准被杠杆化，因为它借入现金（即，T-bills）以投资超过 100%（即， $> 1.0$ ）的市场组合

答案:C

解析:

C 选项正确。C. FALSE: The factors are each significant and the adjusted  $R^2$  is relatively high for empirical finance. In regard to (A), (B) and (D), each is true.

考察: Finance: Portfolio Management      重要程度: [■ ■ □]

科目: Quantitative Analysis

6. The riskfree rate is 5.0% while the current stock price is \$80.00. Two most liquid options both have a strike price of \$70.00 such that the ratio  $K/S(0) = 0.90$ . In addition to the same strike price, both options have the same maturity of three months,  $T = 0.25$  years. The implied volatility of the call is 30.0%, and the implied volatility of the put is 38.0%. What is the trade that exploits this scenario, if any?

无风险利率为 5.0%，当前股票价格为 \$80.00。两种最流动的期权都以 \$70.00 的执行价格，使得比率  $K/S(0) = 0.90$ 。此外，两种期权都具有相同的到期时间，即三个月， $T = 0.25$  年。看涨期权的隐含波动率为 30.0%，看跌期权的隐含波动率为 38.0%。如果存在这样的交易机会，请指出哪种交易策略最合适？

- A. Buy call, sell put, and short the stock  
买入看涨期权，卖出看跌期权，并卖空股票
- B. Buy put, sell call, and buy the stock  
买入看跌期权，卖出看涨期权，并买入股票
- C. Buy the call, buy the put, and short a straddle  
买入看涨期权，买入看跌期权，并卖空跨式期权
- D. Sell the call, sell the put, and buy a straddle  
卖出看涨期权，卖出看跌期权，并买入跨式期权

答案:A

解析:

A 选项正确。True: Buy call, sell put, and short the stock

考察: Derivatives: Options      重要程度: [■ ■ □]

科目: Quantitative Analysis

7. Consider a netting set with two derivative positions: an interest rate swap (IRS) with a marginal CVA of  $-1.450$  and a cross-currency swap (CCS) with a marginal CVA of  $-1.550$ . If the first transaction is the cross-currency swap (CCS) and its incremental CVA is  $-2.100$ , then what is the incremental CVA of the IRS?

考虑一个净额结算集，其中包含两个衍生品头寸：一个利率互换（IRS），其边际 CVA 为  $-1.450$ ，一个交叉货币互换（CCS），其边际 CVA 为  $-1.550$ 。如果第一个交易是交叉货币互换（CCS），其增量 CVA 为  $-2.100$ ，那么利率互换（IRS）的增量 CVA 是多少？

- A. -0.900
- B. -1.100
- C. -1.550
- D. -2.220

答案:A

解析:

A 选项正确。The total CVA is the sum of marginal CVAs and therefore must be  $-1.450 + (-1.550) = -3.000$ . The incremental CVAs must sum to the total (netted) CVA such that, if the IRS is second in sequence, its incremental CVA is equal to  $-3.000 - (-2.100) = -0.900$ .

考察: Credit Risk      重要程度: [■ □ □ □ □]

8. The central feature of Basel III's first pillar is the regulatory capital (i.e., CET1, Tier 1, and total capital) requirements which require minimum fractions of risk-weighted assets (RWA). In addition to these minimums, as of early 2019 (when the buffers were fully phased in), Basel specifies three buffers: the capital conservation buffer (CCB), the G-SIB buffer, and the countercyclical buffer (CCyB). This is all very confusing, so you ask your colleague Mary to summarize these additional buffers and their motivations. She makes the following five points:

- I. For all three, the additional buffer must be CET1
- II. For all three, breach of the buffer requirements implies the bank's ability to pay dividends will be restricted
- III. The CCB requires an additional 2.5% of CET1 capital and is meant to ensure that banks have an additional layer of usable capital that can be drawn down when losses are incurred.
- IV. The additional G-SIB requirement includes five buckets {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} [https://www.bis.org/fsi/fsisummary\\_sib\\_framework.htm](https://www.bis.org/fsi/fsisummary_sib_framework.htm) and is meant to reduce the likelihood and severity of the failure of a global systemically important financial institution
- V. The CCyB requirement varies between zero and 2.5% and is meant to protect the banking sector from periods of excess aggregate credit growth that have often been associated with the build-up of system-wide risks

Is Mary correct in her summary?

巴塞尔 III 第一支柱的核心特征是监管资本（即 CET1、一级资本和总资本）要求，这些要求规定了相对于风险加权资产（RWA）的最低资本比例。除这些最低要求之外，截至 2019 年初（当各类缓冲全部过渡到位时），巴塞尔还规定了三种缓冲：资本留存缓冲（CCB）、全球系统重要性银行缓冲（G-SIB 缓冲）以及逆周期资本缓冲（CCyB）。这些内容非常容易让人困惑，因此你请同事 Mary 总结这些额外缓冲及其设立动机。她给出了以下五点：

- I. 对这三类缓冲而言，额外缓冲部分都必须由 CET1 构成
- II. 对这三类缓冲而言，一旦未满足缓冲要求，银行支付股利的能力将受到限制
- III. CCB 要求额外 2.5% 的 CET1 资本，旨在确保银行拥有一层在发生损失时可以被动用的额外可用资本。
- IV. 额外的 G-SIB 要求包含五个档次 {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} <https://www.bis.org/fsi/fsisummaries/g-sibframework.htm>，旨在降低全球系统重要性金融机构倒闭的可能性及其严重程度
- V. CCyB 要求在零和 2.5% 之间变动，旨在保护银行体系免受总量信贷过度增长时期的影响，而这种过度增长往往伴随着系统性风险的积聚

Mary 在她的总结中是否正确？

- A. No, unfortunately, none of her statements is correct  
不幸的是，她的陈述没有一项是正确的
- B. I. and II. are correct, but the others (III., IV., and V.) are wrong  
I. 和 II. 是正确的，但其他（III.、IV. 和 V.）是错误的
- C. III., IV., and V. are correct, but the others (I. and II.) are wrong  
III.、IV. 和 V. 是正确的，但其他（I. 和 II.）是错误的
- D. Yes, all five statements are correct  
是的，所有五项陈述都是正确的

答案:D

解析:

D 选项正确。 Yes, all five statements are correct

- In regard to true statements (I.) and (II.), the capital conservation buffer (CCB), the G-SIB buffer, and the countercyclical buffer (CCyB) each require common equity Tier 1 (CET1) which is a buffer of the highest quality. Further, breach of any of them implies restrictions on dividend payments (in addition to other supervisor tools such as restriction on executive bonuses)

- In regard to true statement (III.), the BCBS says that “the capital conservation buffer [CCB] was introduced to ensure that banks have an additional layer of usable capital that can be drawn down when losses are incurred” (see [https://www.bis.org/fsi/fsisummaries/b3\\_capital.htm](https://www.bis.org/fsi/fsisummaries/b3_capital.htm)) and GARP says “In the case of the CCB, the rationale roughly follows that for the Prompt Corrective Action (PCA) system built into U.S. capital regulation beginning in 199; i.e., a bank with ratios that begin to approach the minimums should be subject to increasingly stringent supervisory intervention in order to induce a return to well-capitalized status).” –2020 FRM Part II: Operational Risk and Resiliency, 10th Edition.
- In regard to true statement (IV.), the G-SIB requirement includes five buckets {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} and, explains GARP, “In the case of the G-SIB buffer, the rationale is similar to that for the CCB but also recognizes the very large costs to society of distress at G-SIBs (and the higher volatility of losses at some of them). Thus, larger buffers are specified to further reduce the chance of failure.”
- In regard to true statement (V.), “The CCyB [countercyclical capital buffer] has two rationales. One is to provide an instrument for macroprudential restraint of overheating; the other is attentive to the cost of capital,” says GARP. Further, explains BCBS, “The countercyclical capital buffer [CCyB] aims to ensure that banking sector capital requirements take account of the macro-financial environment in which banks operate. Its primary objective is to use a buffer of capital to achieve the broader macroprudential goal of protecting the banking sector from periods of excess aggregate credit growth that have often been associated with the build-up of system-wide risk. Due to its countercyclical nature, the countercyclical capital buffer regime may also help to lean against the build-up phase of the credit cycle in the first place. In downturns, the regime should help to reduce the risk that the supply of credit will be constrained by regulatory capital requirements that could undermine the performance of the real economy and result in additional credit losses in the banking system.” See <https://www.bis.org/bcbs/ccyb/>

Keep in mind the basic requirements of these three additional buffers: the CCB requires +2.5% CET1; the G-SIB requires either {+1.0%, +1.5%, +2.0%, +2.5%, or +3.5%} depending on the bank’s score (see <https://trtl.bz/fsb-g-sibs>); and the CCyB requires between zero and 2.5%.

考察: Finance: Regulatory Capital      重要程度: [■■■□]

科目: Quantitative Analysis

9. A portfolio holds 100,000 shares of a stock and this single position has a value of \$3.0 million. The stock is quoted bid \$29.00, offer \$31.00. The stock’s daily volatility is 1.43% or 143 basis points. For purposes of value at risk (VaR), we will assume the stock’s arithmetic returns are normally distributed (aka, normal VaR) and the expected daily return rounds to zero (under these assumptions absolute VaR is identical to relative VaR). Which is **NEAREST** to the position’s one-day 99.0% confident liquidity-adjusted value at risk (LVar)?

- A. \$15,000
- B. \$80,000
- C. \$200,000
- D. \$1.0 million

答案:C

解析:

**Option C is correct.** The one-day 99.0% VaR is given by  $-0 + 1.43\% * 2.326 * \$3.0 \text{ million} = \$99,800$ .

The liquidity cost (LC) is given by  $(31.00 - 29.00)/30.00 * 0.5 * \$3.0 \text{ million} = \$100,000$ .

The one-day 99.0% VaR is therefore  $99,800 + 100,000 = 199,800$

考察: Market Risk: Liquidity Risk      重要程度: [■■■□]

科目: Quantitative Analysis

10. According to Grinold, typical or illustrative information ratios (IR) include: 0.500 is good, 0.750 is very good and 1.00 is exceptional. In regard to risk aversion,  $\lambda$ , 0.150 is restrained, 0.100 is moderate, and 0.050 is aggressive. Patricia is a manager

with a “very good” information ratio of 0.750 and a “moderate” risk aversion of 0.10. If she shifts from this moderate risk aversion to an aggressive risk aversion of 0.050, what is the impact on the optimal level of active risk?

- A. No effect
- B. Optimal active risk reduces by one-half
- C. Optimal active risk Increases by about +41%
- D. Optimal active risk doubles

答案:D

解析:

**D 选项正确。** True: Optimal active risk doubles As the optimal level of active risk is given by  $IR/(2 * \lambda)$ , Patricia doubles her optimal level when she goes from  $0.750/(2 * 0.10) = 3.750$  to  $0.750/(2 * 0.050) = 7.500$

考察: Portfolio Management: Fundamental Law of Active Management      重要程度: [■■■]

科目: Quantitative Analysis

**11.** For his term structure model, Peter considered the Vasicek model, but on reflection, he prefers the Cox-Ingersoll-Ross (CIR). In regard to the CIR, which of the following statements is TRUE?

- A. A drawback of the CIR is that it does not mean revert to a long-term rate
- B. In the CIR, the drift is time-dependent such that CIR is a no-arbitrage model
- C. An advantage of the CIR is that the terminal distribution of the short rate is eventually (e.g., > 10 years) normally distributed
- D. Unlike the Vasicek model’s constant volatility, the CIR’s basis point volatility is not constant, and this ensures the short rate cannot become negative

答案:D

解析:

**D 选项正确。** True: Unlike the Vasicek model’s constant volatility, the CIR’s basis point volatility is not constant, and this ensures the short rate cannot become negative.

In regard to (A), (B), and (C), each is FALSE.

考察: Term Structure Models      重要程度: [■■■]

科目: Quantitative Analysis

**12.** Jason is a small business owner specializing in the wholesale of steel. Jason worked hard on his business, so he purchased insurance to help safeguard his company on January 4th, 2023. Later that month, on January 31st, he made a significant payment to secure a shipment of steel that will arrive on February 2nd, 2023. Jason delivered the steel to two clients on February 8th and 9th; however, he did not receive payment from the clients until February 13th. After receiving the proceeds Jason deposited the money at ABC bank.

Among these transaction dates, on which did Jason increase his credit risk?

- A. None of the transactions increase his credit risk
- B. His credit risk increased on the Jan dates (Jan 4th and 31st) but was unchanged on Feb dates (Feb 8th, 9th and 13th)
- C. His credit risk increased on the Feb dates (Feb 8th, 9th, and 13th) but was unchanged on Jan dates (Jan 4th and 31st)
- D. His credit risk increased on all of the cited transaction dates; i.e., both Jan dates (4th and 31st) and all Feb dates (8th, 9th, and 13th)

答案:D

解析:

D 选项正确。 True: His credit risk increased on all of the cited transaction dates; i.e., both Jan dates (4th and 31st) and all Feb dates (8th, 9th, and 13th)

- Jan 4th: Purchasing insurance increases credit risk because Jason is buying a policy that will not be delivered if ever once an event occurs. *Contingent claims* are a type of credit risk. According to the text, “A sixth type of transaction is a special case of a claim on an asset—a contingent claim. The claim is contingent on certain events occurring, such as a loss covered by an insurance policy. At policy inception, the policyholder has no claim on the insurer. However, once the insured suffers a covered loss, the insured has a claim. If the insurer fails to pay the claim, this would constitute a credit loss.” (†)
- Jan 31st: He prepays for an order and thus has the possibility of his counterpart not delivering on their obligation. According to the text, “*Prepayment of goods and services* is a fourth type of transaction that involves credit risk. Delivery is expected at a certain time and of a certain quality and/or performance, and the failure of the counterparty may lead to the loss of the advanced payments and also generates business interruption costs.” (†)
- February 8th and 9th: Jason delivered an order of steel to his clients. Either of his clients could have not delivered on their obligation to pay him, and thus, Jason increased his credit risk. According to the text, “The third type is the sale of a product or a service without immediate cash payment. The seller sends an invoice to the buyer after the product has been shipped or the service performed, and the buyer has a few weeks to pay. This is known as an *account receivable*.” (†)
- February 13th: He deposited the proceeds of his sale and his earnings from last quarter at ABC Bank. Jason granted a third-party custody over his assets, and thus, he has the risk of the bank not being able to repay his assets. Per the text, “A fifth type of transaction that creates credit risk involves a party’ s claim on an asset in the custody of or under the management of another party, such as a bank deposit.” (†)

Credit risk can be generated in myriad ways. The text(†) lists seven:

1. Lending: Risk of non-repayment from the borrower to the lender.
2. Leases: Risk that the lessee will fail to make future payments for the use of an asset.
3. Accounts Receivable (Sale of Product/Service without Immediate Payment): Risk that the buyer will not pay after receiving the product or service.
4. Prepayment of Goods and Services: Risk that the counterparty will not deliver or meet the quality/performance after receiving payment.
5. Deposits and Bank Balances: Risk that the financial institution holding deposits or managing assets will default.
6. Contingent Claims (e.g., Insurance, Pension Fund Claims): Risk that the insurer or sponsor will not be able to pay out the claim or pension liabilities.
7. Derivative Exposures: Risk from future payment obligations in derivative transactions like swaps or futures, which are dependent on the underlying asset’ s market value.

(†) Sylvain Bouteille and Diane Coogan-Pushner, **The Handbook of Credit Risk Management: Originating, Assessing, and Managing Credit Exposures** (2nd Edition, Hoboken, NJ: John Wiley & Sons, 2022).

考察: Finance: Credit Risk    重要程度: ☒☒☐

科目: Quantitative Analysis

13. Peter is studying for the Financial Risk Manager (FRM) exam, and he thinks the historical evolution of the Basel regulations is confusing. In particular, he wants to better understand the difference between the original Basel I accord (aka, the 1988 Basel Accord that was implemented by 1992) and Basel II, which was finally published over ten years later in 2004. His colleague Mary explains that, in comparison to the original accord, Basel II contained four significant innovations. Specifically, in comparison to the original Basel I accord (which do include the 1995/1996 Amendments) that came before, Mary says that significant innovations in Basel II included the following:



- I. In addition to credit risk and market risk, Basel II required capital for operational risk
- II. Basel II eliminated risk weights and risk-weighted assets (RWA) and replaced them with direct calculation of risk charges
- III. Basel II contained specific requirements for supervision related to capital and risk management (Pillar 2) and required public disclosures (Pillar 3)
- IV. To fine-tune the accord’ s design, Basel II made repeated use of Quantitative Impact Studies (QIS) to which banks contributed data (i.e., feedback) that was analyzed by supervisors

Is Mary correct?

- A. Yes, all four statements are correct
- B. I. and II. are correct but III and IV are inaccurate
- C. Only II. is inaccurate but I, III, and IV are correct
- D. Unfortunately, none of her statements are correct

答案:C

解析:

**Option C is correct.** Basel II did NOT eliminate risk-weighted assets (or risk weights); risk-weighted assets (RWA) do continue in Basel III/IV. According to GARP’ s Mark Carey, the four significant innovations of Basel II (as an improvement over the original Basel I accord) were: more sophisticated risk weight formulas; the addition of an operational risk capital charge; Pillars 2 and 3; and the repeated use of the Quantitative Impact Studies (QIS). Many experts consider the additional Pillars to be a highly significant development. As Carey explains, “The three pillars represented a push toward convergence of national practices [especially in the context of the “rules versus principles” disparity]. Specifically, Pillar 2 mandated that supervisors require banks to have more than the minimum amount of capital as well as internal capital adequacy and assessment processes (ICAAP) that take their risk profile into account ... National discretion regarding enforcement of the accord’ s provisions was reduced, and national regulators were to be transparent about their implementation efforts, including those concerning the requirements in excess of the minimums. Pillar 3 required more qualitative and quantitative disclosures, in the hope that pressure from market participants would help improve banks’ practices [aka, market discipline].” (Source: 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition. Pearson Learning Solutions)

考察: Finance: Basel Accords      重要程度: ■■□

科目: Quantitative Analysis

**14.** Assume the spot foreign exchange (FX) rate between the US dollar and the Japanese Yen is 110.00 which can be represented as USDJPY 110.00 or JPY 110.00 where USD is the base currency and JPY is the quote currency; aka, quote 110.00 yen per one dollar of the base or “per unit” currency. The USD interest rate is 2.50% and the JPY interest rate is 0.450%. The period is one year, and the compound frequency is annual. If the covered interest rate (CIP) arbitrage framework enforces a perfectly accurate one-year forward exchange rate, then what is the implied one-year “swap rate” which is here simply defined as the difference between the forward FX rate and the spot FX rate; put simply, what is the difference,  $F(\text{USDJPY}) - S(\text{USDJPY})$ , or  $F(\text{USDJPY}) - 110.00$ ?

- A. -2.20 JPY (or -2.20 USDJPY)
- B. Zero
- C. + 5.80 JPY
- D. + 13.70 JPY

答案:A

解析:

**A 选项正确。** True:  $F - S = -2.20$  USDJPY

If CIP is enforced, then  $F - S = S \times [(1 + \text{Rf\_JPY})/(1 + \text{Rf\_USD}) - 1] = 110.00 \times [(1 + 0.45\%)/(1 + 2.50\%) - 1] = -2.20$ .

考察: Financial Markets and Products      重要程度: [■■■□]

科目: Quantitative Analysis

**15.** Volatility is an important risk factor according to Andrew Ang. The strong negative relationship between VIX changes and stock returns is explained by (at least) two dynamics. The first is called the leverage effect. The “second channel is a time-varying risk premium story and is the one the basic CAPM [capital asset pricing model] advocates,” explains Ang.(†)

Assume the riskfree rate is 3.0% while the equity risk premium (EPR; aka, the market’s expected excess return) is 6.0% because the market’s expected return is 9.0%. According to the CAPM, and consistent with Ang’s theory on the transmission channel between volatility and stock returns, which of the following should cause a DECLINE in a stock’s price?

- A. The VIX Index declines from 21.00 to 19.00
- B. The market’s volatility decreases and the stock’s volatility decreases along with it
- C. The stock’s volatility decreases, which corresponds to an increase in its leverage via the leverage effect
- D. The stock’s volatility increases (along with an increase in its correlation to the market), while the market’s volatility is unchanged

答案:D

解析:

**Option D is correct.** The stock’s volatility increases (along with an increase in its correlation to the market), while the market’s volatility is unchanged.

An increase in the stock’s volatility and/or increase in its correlation to the market, while the market’s volatility is unchanged (aka, ceteris paribus), implies an increase in the stock’s beta. This increases the stock’s expected return. Consequently, the stock’s price should decline.

Ang (emphasis ours): “The correlation between VIX changes and stock returns is  $-39\%$ , so stocks do badly when volatility is rising. The negative relation between volatility and returns is called the leverage effect. When stock returns drop, the financial leverage of firms increases since debt is approximately constant while the market value of equity has fallen. This makes equities riskier and increases their volatilities. There is another channel where high volatilities lead to low stock returns: **an increase in volatility raises the required return on equity demanded by investors, also leading to a decline in stock prices.** This second channel is a time-varying risk premium story and is the one that the basic CAPM advocates: as market volatility increases, discount rates increase and stock prices must decline today so that future stock returns can be high.” (†)

In regard to (A), (B), and (C), each is false.

(†) Andrew Ang, Asset Management: A Systematic Approach to Factor Investing (New York: Oxford University Press, 2014).

考察: Finance: CAPM      重要程度: [■■■□]

科目: Quantitative Analysis

**16.** An analyst fits a Vasicek model when the current short-term rate is 4.60%. If the theta,  $\theta$ , parameter is 13.50%, then each of the following statements is true EXCEPT which is false?

- A. A key advantage of the Vasicek model is that it is a relatively simple model
- B. If the half-life is nine years, then we might expect the short rate to reach  $\sim 9.0\%$  in nine years or so
- C. The standard deviation of the short rate in ten years is less than it would be without mean reversion
- D. Because it is a multi-factor model, the Vasicek is popularly applied to hedge portfolios that contain short-, medium- and long-term bonds

答案:D

解析:

**D 选项正确.** False. Vasicek is a single-factor model; to hedge short-, medium- and long-term bonds would require, for example, the three-factor Gauss+ model.



In regard to (A), (B), and (C), each is TRUE.

考察: Fixed Income: Vasicek Model      重要程度: [■□□]

科目: Quantitative Analysis

17. Firm A, Firm B, and Firm C are three companies in the bottled water industry. All companies have a mix of debt and equity in their capital structure. In 2022 all three companies had credit losses that lowered EBIT. Below is a summary of their financial information for the year 2022.

|                         | Firm A      | Firm B      | Firm C      |
|-------------------------|-------------|-------------|-------------|
| Total Assets            | \$5,500,000 | \$5,000,000 | \$3,900,000 |
| Total Debt              | \$2,500,000 | \$1,000,000 | \$1,000,000 |
| Debt Interest Rate      | 3.40%       | 3.40%       | 3.40%       |
| Total Equity            | \$3,000,000 | \$4,000,000 | \$2,900,000 |
| Provision for Loan Loss | \$50,000    | \$15,000    | 1,000       |
| EBIT                    | \$75,000    | \$75,000    | \$75,000    |
| Tax Rate                | 21%         | 21%         | 21%         |

Given the above information, which firm has the higher return on equity (ROE)?

- A. Firm C has the highest ROE due to its lower equity position.
- B. Firm A had the higher ROE because of its large debt position.
- C. Without credit losses, firm A would have had the highest ROE.
- D. Without credit losses, firm B would have had the highest ROE.

答案:A

解析:

A 选项正确。 True: Firm C has the highest ROE due to its lower equity position.

- Firm A has a negative ROE (−0.33%) because of its large debt position (as illustrated below). Even after sustaining credit losses the firm still must repay its debt holders. The A sustained heavy credit losses that lowered EBIT. However, even after adjusting for its credit losses Firm A still would have had a smaller ROE than firms B and C.
- Firm B had a positive ROE, but the ROE is very small because of its large equity compared to firms C and A. Firm A sustained a large credit loss that lowered EBIT. However, even after adjusting for its credit losses the firm would have still had a smaller ROE than firm C.
- Firm C had the highest ROE because of its small equity and modest credit loss.

|                            | Firm A        | Firm B      | Firm C      |
|----------------------------|---------------|-------------|-------------|
| Total Assets               | \$5,500,000   | \$5,000,000 | \$3,900,000 |
| Total Debt                 | \$2,500,000   | \$1,000,000 | \$1,000,000 |
| Debt Interest Rate         | 3.40%         | 3.40%       | 3.40%       |
| Total Equity               | \$3,000,000   | \$4,000,000 | \$2,900,000 |
| Provision for loan loss    | \$50,000      | \$15,000    | 1,000       |
| EBIT                       | \$75,000      | \$75,000    | \$75,000    |
| Tax Rate                   | 21%           | 21%         | 21%         |
| <b>Solution</b>            |               |             |             |
| EBIT                       | \$75,000      | \$75,000    | \$75,000    |
| Interest                   | \$85,000.00   | \$34,000.00 | \$34,000.00 |
| Tax                        | 0             | \$8,610.00  | \$8,610.00  |
| NI                         | (\$10,000.00) | \$32,390.00 | \$32,390.00 |
| ROE                        | -0.33%        | 0.81%       | 1.12%       |
| <b>Without Credit Loss</b> |               |             |             |
| EBIT                       | \$125,000     | \$90,000    | \$76,000    |
| Interest                   | \$85,000.00   | \$34,000.00 | \$34,000.00 |
| Tax                        | \$8,400.00    | \$11,760.00 | \$8,820.00  |
| NI                         | \$31,600.00   | \$44,240.00 | \$33,180.00 |
| ROE                        | 1.05%         | 1.11%       | 1.14%       |

考察: Finance: Financial Statement Analysis      重要程度: [■■■□]  
 科目: Quantitative Analysis

18. The three lines of defense model (3LoD) has existed in some variation for over 30 years, although it achieved more visibility in 2013 when the Institute of Internal Auditors (IIA) published its initial position paper on the 3LoD, which it updated in 2021 (The IIA’s Three Lines Model, an update of the Three Lines of Defense). GARP recommends firms refer to this update because the IIA “has refined the model to reflect the evolution of risk management within organizations and aim to foster closer collaboration between business functions including internal audit.”

In regard to the 3LoD model, each of the following is true EXCEPT which is false?

- A. The first line defines the organization’s risk appetite but should not attempt to directly influence its risk culture
- B. The first line is responsible for implementing and maintaining effective controls to manage material operational risks
- C. The second line is part of management (and owns the ORM methodology) but provides support to the first line in order to help the firm create and protect value
- D. The third line has the clearest boundaries due to its independence

答案:A  
 解析:

A 选项正确。A. False, both clauses. Instead, the following is true: The risk appetite (i.e., risk appetite statement) is developed by senior management and approved by the board. The risk culture is not the sole responsibility but is foundational to the firm’s actual ORMF, such that all employees bear some responsibility for culture.

In regard to (B), (C), and (D), each is TRUE.

考察: Operational Risk      重要程度: [■■■□]  
 科目: Quantitative Analysis

19. Liquidity Early Warning Indicators (EWI) can be compared to an automobile’s dashboard signal system where the appearance of a red light points our attention to something that may become a problem if not addressed. Each of the following is a credible red indicator (aka, red flag) in a Liquidity Early Warning Indicator (aka, Liquidity EWI) dashboard EXCEPT which is not a red (red flag) liquidity indicator?

- A. Narrowing debt/CDS spreads
- B. Sudden increase in debt costs
- C. Sudden growth in assets accompanied by volatile liabilities
- D. Rapid decline in the weighted average maturity of liabilities

答案:A

解析:

**A 选项正确。** A. FALSE. Instead, the liquidity EWI is: a widening debt/CDS spreads (consistent with a sudden increase in debt costs)

In regard to (B), (C), and (D), each is a candidate Liquidity EWI, per the Study Note: “The Basel Committee on Banking Supervision (BCBS) recommends certain measures as EWI. These measures include [collapsed here, new emphasis ours]:

- An unusual growth in assets, particularly when accompanied by volatile liabilities;
- Debt (credit) spreads *widen*, and/or credit default swap (CDS) spreads *widen*.
- Declining diversity in the makeup of assets and liabilities; Growing currency mismatches;
- When the weighted average of liabilities’ maturity declines;
- Positions going beyond or getting close to regulatory limits;
- Certain product line experience negative trends; The financial condition of the bank weakens; Public press that is negative;
- A downgrade in the credit rating; A decline in the stock price;
- Debt costs increase; Retail and/or wholesale funding costs increase;
- Counterparties becoming nervous about the financial condition of the bank;
- Credit lines are lowered;
- Outflows of retail deposits at an increased pace; Certificates of deposit (CDs) are increasingly redeemed;
- Longer-term funding opportunities become more difficult; and Placing short-term liabilities becomes more difficult”

考察: Liquidity Risk Management      重要程度: [■ ■ □]

科目: Quantitative Analysis

**20.** In factor theory, an asset’s risk premium (aka, expected excess return) is given by  $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$  where the asset’s beta is with respect to the stochastic discount factor (SDF; aka, pricing kernel) and  $(m)$  in an index of bad time. Let’s make the following assumptions about an asset:

- The riskfree rate is 3.0% such that  $E(m) = 1/1.03 = 0.9709$
- The variance( $m$ ) equals  $30.0\%^2 = 0.090$
- The covariance( $r_i, m$ ) equals  $-0.10680$

Which is **nearest** to the asset’s risk premium which refers to the asset’s expected excess return,  $E(r_i) - R_f$ ?

- A. -5.0%
- B. +4.0%
- C. +8.0%
- D. +11.0%

答案:D

解析:

**D 选项正确。** The expected excess return,  $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$  where  $\beta(i, m) = \text{cov}(r_i, m) / \text{var}(m)$  and  $\lambda(m) = -\text{var}(m) / E(m)$ .

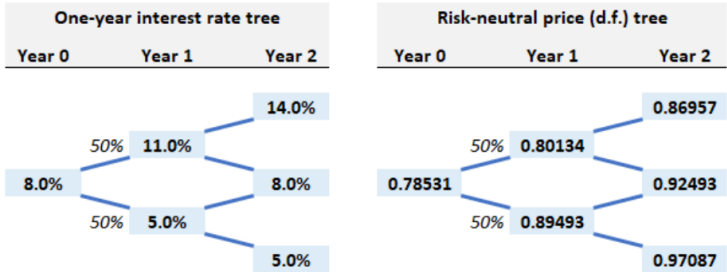
In this case,  $\beta(i, m) = \text{cov}(r_i, m) / \text{var}(m) = -0.10680 / 0.090 = -1.186667$ , and  $\lambda(m) = -\text{var}(m) / E(m) = -0.090 / 0.9709 = -0.0926975$ .

Therefore,  $E(r_i) - R_f = \beta(i, m) \times \lambda(m) = -1.186667 \times -0.0926975 = 0.1100011$  or about 11.0%.

考察: Finance: Factor Theory    重要程度: [■■■]

科目: Quantitative Analysis

**21.** Below (in the left-hand panel) is an interest rate tree for the one-year rate where the initial rate of 8.0% jumps 300 basis points to either 11.0% or 5.0% next year. In two years, the one-year rate will be either 2.0%, 8.0%, or 14.0%. Next, we will assume a three-year zero-coupon bond. Without a risk premium, for example, the date 0, state 2 price would be  $0.877193 = 1 / (1 + 14\%)$ . However, we will instead assume a risk premium. Specifically, investors require compensation of 50 basis points for each year of duration risk. The right-hand panel displays the associated risk-neutral (i.e., , assumes the risk premium) price tree for this three-year zero-coupon bond



In regard to the three-year zero-coupon bond, which of the following is true?

- A. The expected return over the next year is 7.50%
- B. The expected return over the next year is 9.00%
- C. The implied 3-year spot rate term structure is downward sloping (aka, inverted)
- D. Jensen’s inequality implies that expected excess return over two years is less than 50 basis points

答案:B

解析:

**B 选项正确。** True: The expected return over the next year is 9.00%.

We don’ t require the formula. The risk premium is 50 basis points for each year of duration risk, and the three-year bond creates two years of duration risk. The expected return, therefore, is 100 basis points above 8.0%.

考察: Fixed Income    重要程度: [■■■]

科目: Quantitative Analysis

**22.** Old Crow Bank had a credit committee meeting yesterday, and the following is a copy of the notes taken during the meeting:

- All members are present Jan (Chief Risk Officer), Eddie (Head of Compliance), Alyssa (Head of Legal), Don (Head of Tax), and Ashton (Chief Financial Officer).
- Jan initiated the meeting by confirming if everyone had received an email containing approval documents from credit officer Doug, who had originated the loan. Everyone confirmed receiving the email with the documents yesterday.
- Jan provided an overview of the loan for approximately five minutes, after which she proceeded to inquire about the opinions of those present. It was the consensus of all individuals, except for Alyssa, that the loan be approved. Alyssa expressed concern regarding the trustworthiness of the counterparty.

- Jan respects Alyssa’s opinion but disagrees and initiates a vote. Two other committee members agree with Jan, resulting in loan approval.

What is the most probable issue with the credit committee’s meeting?

- A. The approval documents should have been distributed earlier.
- B. Jan monopolized the conversation and did not properly facilitate discussion.
- C. The meeting was not properly documented.
- D. The vote was not unanimous, so the loan should have been rejected.

答案:C

解析:

**Option C is correct.** In this scenario, all the approval documents were appropriately distributed to the committee before the meeting. Jan did spend time explaining the investment to the committee, but she also helped facilitate discussion among the committee members and listened to everyone’s opinion. Votes do not have to be unanimous, but they should be held unless the decision is unanimous. However, meetings should be accurately documented in case an investment goes awry. In the case of a bad transaction, auditors would want to know what was discussed in the meeting and if the reasoning is behind the investment decision.

考察: Credit Risk Management      重要程度: [■ ■ □]

科目: Quantitative Analysis

**23.** A bank is using the VaR and stressed VaR (SVaR) market risk framework according to the Basel 2.5 guidelines. The bank’s internal models for market risk have generated the following risk measures for the current trading book positions (VaR and SVaR are in USD millions):

| Multiplicative factors, m and m(S) |              |         |                | 3.0     |
|------------------------------------|--------------|---------|----------------|---------|
| Confidence Level                   | Previous day |         | 60-day average |         |
|                                    | VaR          | SVaR    | VaR            | SVaR    |
|                                    | (mm)         | (mm)    | (mm)           | (mm)    |
| 95.00%                             | \$110.0      | \$190.0 | \$127.0        | \$238.0 |
| 99.00%                             | 185.0        | 360.0   | 140.0          | 270.0   |
| 99.90%                             | 283.0        | 740.0   | 228.0          | 490.0   |

The supervisor has set both multiplication factors, m and m(S), to 3.0. What is the correct capital requirement for general market risk for the bank under Basel 2.5? (note: this question inspired by Question 42 of GARP’s 2020 Part 2 Practice Exam).

- A. \$630.0 million
- B. \$810.0 million
- C. \$1,230.0 million
- D. \$2,154.0 million

答案:C

解析:

**C 选项正确。** True: \$1,230.0 million

The Basel 2.5 market risk capital requirement requires a 99.0% confidence level and is given by:

$$MRC = \max[VaR(t - 1), m \times VaR(60\text{-day avg})] + \max[SVaR(t - 1), m(S) \times SVaR(60\text{-day avg})]$$

In this case,

$$MRC = \max[185.0, 3.0 \times 140.0] + \max[360.0, 3.0 \times 270.0] = \$1,230.0 \text{ million.}$$

考察: Market Risk      重要程度: [■■■]

科目: Quantitative Analysis

24. If a corporation’s marginal tax rate is 30.0% and it purchases an A-rated municipal bond that offers a 4.20% gross yield (aka, yield to maturity), what is the tax equivalent yield (TEY)?

- A. 2.45%
- B. 3.50%
- C. 6.00%
- D. 16.67%

答案:C

解析:

C 选项正确。 TRUE: 6.00%.  $TEY = \text{after-tax return on tax-exempt investment} / (1 - \text{marginal tax rate})$ ; in this case,  $TEY = 4.20\% / (1 - 0.30) = 6.00\%$

考察: Finance: Fixed Income      重要程度: [■■■■]

科目: Foundations of Risk Management

25. Andrew Ang introduces factors and divides them primarily into two types: macro, fundamental-based factors versus investment-style factors. He writes, “Factors drive risk premiums. One set of factors describes fundamental, economywide variables like growth, inflation, volatility, productivity, and demographic risk. Another set consists of tradeable investment styles like the market portfolio, value-growth investing, and momentum investing.” (†)

Ang also claims that the three most important macro factors are growth, inflation, and volatility. His evaluation of these macro factors is based on a long-term historical sample (specifically, 1952:Q1 to 2011:Q4). It is important to qualify the sample window because we cannot be sure that past is prologue; e.g., interest rates experienced two long-term secular trends during this window.

In regard to his historical analysis, each of the following statements is true EXCEPT which is false?

- A. Government and investment-grade bonds performed BETTER during economic recessions than during expansions
- B. Both large and small (cap) stocks perform significantly BETTER during economic expansions than during recessions
- C. During periods of high inflation, all five asset classes perform significantly BETTER than during periods of low inflation
- D. All five assets classes are much MORE VOLATILE during recessions (or periods of low GDP growth) than during expansions

答案:C

解析:

C 选项正确。 False. To be true, this should instead read: During periods of high inflation, all five asset classes perform significantly WORSE than during periods of low inflation.

In regard to (A), (B) and (D), each is TRUE.

考察: Factor Investing      重要程度: [■■■]

科目: Financial Markets and Products

26. Below is the price tree for a stylized constant-maturity Treasury (CMT) swap struck at 2.0% with a notional of \$1.0 million. The risk-neutral probabilities are displayed, specifically: 80.09% (d = 19.91%) from date 0 to date 1; and 65.20% (d = 34.80%) from date 1 to date 2. Using these risk-neutral probabilities returns an expected discounted value (EDV) of \$3,702.11 for the value of the CMT swap. Please notice this exactly matches Tuckman’s Figure 7.8 in his 4th edition value. This is the so-called model value.

|                            |                         |               |
|----------------------------|-------------------------|---------------|
| Notional                   | \$1,000,000             |               |
| Strike rate                | 2.00%                   |               |
| <u>i-rate evolution</u>    | <u>T= 0.5</u>           | <u>T= 1.0</u> |
|                            |                         | 3.00%         |
|                            | 2.50%                   | 2.00%         |
| 2.0%                       | 1.50%                   | 1.00%         |
| Risk-neutral               | 80.09%                  | 65.20%        |
| Probabilities:             | 19.91%                  | 34.80%        |
| <u>Price tree: 2 dates</u> | <u>T= 1.0</u>           |               |
|                            |                         | \$5,000.00    |
|                            | \$5,719.52              | \$0.00        |
| \$3,702.11                 | -\$4,227.29             | -\$5,000.00   |
| \$3,694.81                 | ← Observed Market Price |               |
| 0.25%                      | ← OAS                   |               |

Assume the observed market price of this CMT swap is \$3,694.81 (as highlighted in purple), which is \$7.30 less than the market price. This implies an option-adjusted spread (OAS) of 25 basis points. Which of the following is the **best** interpretation of this OAS?

- A. The model returns the market price if we adjust the risk-neutral probabilities to 55.09% and 44.91%
- B. The model returns the market price if we assume adjusted cash flows; e.g., \$3,750 instead of \$5,000 on (state 2, date 2)
- C. The model returns the market price if we use discount rates of 2.25% (date 0, state 0), 1.75% (date 1, state 0), and 2.75% (date 1, state 1)
- D. We cannot interpret the OAS here because this is a CMT swap, and the OAS requires an embedded option; e.g., a bond with an embedded call option

答案:C

解析:

**C 选项正确。** True: The model returns the market price if we use discount rates of 2.25% (date 0, state 0), 1.75% (date 1, state 0), and 2.75% (date 1, state 1)

考察: Fixed Income Analysis      重要程度: [■■■]

科目: Quantitative Analysis

**27.** Strongcore Financial Corporation has purchased a long-term at-the-money (ATM) call option from a risky counterparty. The option's stock and strike price are \$25.00 million. The underlying asset's volatility,  $\sigma$ , is estimated to be 20.0% while the risk-free rate is 3.0% per annum. The European-style option matures in 3.0 years. In regard to the standard normal cumulative distribution function, in this scenario  $N(d1)$  equals 0.66750 and  $N(d2)$  equals 0.53450.



With respect to its counterparty risk, Strongcore assigns a hazard rate estimate of 6.0% and believes its counterparty could default at any time during the three-year option period; i.e., the cumulative default probability is relevant for this approximation. Further, if its counterparty defaults, Strongcore assigns a conservative recovery rate estimate of 30.0%. In summary, the assumptions are:  $S = K = \$25.0$  million,  $\sigma = 20.0\%$ ,  $R_f = 3.0\%$ ,  $T = 3.0$  years;  $\lambda$  (i.e., hazard rate) = 6.0%, and  $LGD = 70.0\%$ .

In this situation, Strongcore does NOT want to compute the standard credit value adjustment (CVA) formula which is the summation of discounted expected exposures,  $EE(t)$ , over multiple future periods. Instead, Strongcore wants to rely on the simpler “semi-analytical method” that produces an estimate as a function of the position’s current value, the cumulative default probability,  $PD(0,T)$ , and the loss given default,  $LGD$ . Under this simpler approximation, which of the following is **nearest** to the CVA charge for the long option position?

- A. \$15,600
- B. \$90,000
- C. \$516,000
- D. \$1.1 million

答案:C  
解析:  
C 选项正确。 \$516,000

$CVA = -LGD \times PD(0,T) \times V(t)$ . In this case,  $V(t) = S \times N(d1) - K \times \exp(-rt) \times N(d2) = 25 \times 0.66750 - 25 \times \exp(-0.030 \times 3) \times 0.53450 = \$4.475$  million. The cumulative  $PD = 1 - \exp(-0.060 \times 3) = 16.473\%$ . Therefore the CVA approximation is given by  $70\% \times 16.473\% \times 4.475 \text{ million} = \$0.5160 \text{ million}$ , or \$516,002.

考察: Credit Risk      重要程度:  
科目: Quantitative Analysis

**28.** Andrew is preparing to make a presentation to the board. The presentation’s goal is to introduce a new, proposed operational risk management (ORM) framework and get the board’s feedback on the way to eventual approval. It happens to be the case that his company is publicly traded with a high cost of capital due to its above-average beta. The board members are generally more familiar with market and credit risk but less familiar with operational risk. Therefore, his presentation will review several of the features (and nature) of operational risk, including its dynamic evolution. His presentation will briefly explore the skewed, heavy-tailed feature of operational risk. For candidate frequency distributions, he will illustrate the Poisson and negative binomial. For candidate severity distributions, he will illustrate the log-normal and generalized Pareto Distribution (GPD).

In addition to these features (i.e., dynamic, heavy-tailed), which of the following is TRUE about operational risk?

- A. The easiest way for a high-beta firm to lower its beta is to centralize the operational risk management (ORM) function
- B. Operational risk is highly homogeneous because so-called boundary events define clusters that contain loss events and their shared (aka, common) features
- C. ORM is valuable because it reduces regulatory capital, and regulatory capital is a good indicator of the firm’s quality of risk management
- D. Most causes of operational risk trace to control weaknesses; human biases and behavioral failings; and/or operating environment changes

答案:D  
解析:

**Option D is correct.** Most operational risk causes trace to control weaknesses; human biases and behavioral failings; and/or operating environment changes  
In regard to (A), (B), and (C), each is FALSE.

- Operational risk is predominantly idiosyncratic and diffuses within the firm. Beta is a systematic risk.

- Operational risk is highly heterogeneous but interconnected. Further, boundary events are “risk events that materialize in a different category than they cause” -Source: GARP Introduction to Operational Risk and Resilience, Section 1.5)
- According to GARP, “regulatory operational risk capital is not sensitive enough to a firm’s risk profile and thus brings no real incentive to optimize the management of operational risks. In fact, regulatory capital is often a poor indicator of risk management quality.” -Source: Chapter (GARP Introduction to Operational Risk and Resilience, Section 1.7)

考察: Operational Risk     重要程度: [■ ■ □]

科目: Quantitative Analysis

29. Over the next 24 hours, Greenlux State Bank estimates that the following cash inflows and outflows (all figures in millions) will occur:

|                                           |         |
|-------------------------------------------|---------|
| Deposit withdrawals                       | \$70.0  |
| Deposit inflows                           | \$100.0 |
| Scheduled loan repayments                 | \$60.0  |
| Acceptable loan requests                  | \$90.0  |
| Borrowings from the money market          | \$80.0  |
| Sales of bank assets                      | \$30.0  |
| Stockholder dividend payments             | \$20.0  |
| Revenues from sale of nondeposit services | \$10.0  |
| Repayment of bank borrowings              | \$50.0  |
| Operating expenses                        | \$40.0  |

What is the bank’s projected net liquidity position?

- A. -30.0 million
- B. +10.0 million
- C. +40.0 million
- D. +90.0 million

答案:B

解析:

**B. TRUE: +10.0 million**

Inflows include: Deposit inflows = 100+

Scheduled loan repayments = 60+

Borrowings from the money market = 80+

Sales of bank assets = 30+

Revenues from sale of nondeposit services = 10;

for total inflows of  $\$100 + \$60 + \$80 + \$30 + \$10 = \$280.0$  million.

Outflows include:

Deposit withdrawals = 70+

Acceptable loan requests = 90+

Stockholder dividend payments = 20+

Repayment of bank borrowings = 50+

Operating expenses = 40;

for total outflows of  $\$70 + \$90 + \$20 + \$50 + \$40 = \$270.0$  million

Therefore, the projected net liquidity position =  $\$280 - 270 = +10.0$  million.

考察: Finance      重要程度: [■ ■ □]

科目: Quantitative Analysis

**30.** Emily manages a \$20.0 million portfolio allocated into two positions (aka, two components): a Technology ETF, and a Real Estate Investment Trust (REIT) ETF. She compares four different allocations, where the weight assigned to the Tech ETF component is either 35.0%, 50.0%, 65%, or 80.0%.

|                             | Mix (Allocation) Scenarios |          |          |          |
|-----------------------------|----------------------------|----------|----------|----------|
|                             | #1                         | #2       | #3       | #4       |
| Weight in Tech ETF (%)      | 35.0%                      | 50.0%    | 65.0%    | 80.0%    |
| Position (000s)             | \$7,000                    | \$10,000 | \$13,000 | \$16,000 |
| Individual VaR (000s)       | \$2,303                    | \$3,290  | \$4,277  | \$5,264  |
| Marginal VaR                | 0.020                      | 0.118    | 0.231    | 0.305    |
|                             |                            |          |          |          |
|                             | #1                         | #2       | #3       | #4       |
| Weight in REIT ETF (%)      | 65.0%                      | 50.0%    | 35.0%    | 20.0%    |
| Position, USD (000s)        | \$13,000                   | \$10,000 | \$7,000  | \$4,000  |
| Individual VaR (000s)       | \$6,415                    | \$4,935  | \$3,454  | \$1,974  |
| Marginal VaR                | 0.461                      | 0.386    | 0.232    | 0.039    |
| Undiversified Portfolio VaR | \$8,718                    | \$8,224  | \$7,731  | \$7,237  |

If her goal is to minimize the portfolio’ s diversified value at risk (VaR), which of the above allocations is **best**?

- A. Mix #1 with 35.0% allocated to the Tech ETF
- B. Mix #2 with 50.0% allocated to the Tech ETF
- C. Mix #3 with 65.0% allocated to the Tech ETF
- D. Mix #4 with 80.0% allocated to the Tech ETF

答案:C

解析:

**Option C is correct.** Mix #3 with 65.0% allocated to the Tech ETF.

Risk is minimized (i.e., the portfolio’ s diversified VaR is minimized) when the marginal VaRs are constant. Among the four mixes, mix #3 is clearly the choice where the marginal VaRs are nearest (i.e., 0.231 versus 0.232 is almost an exact match). Mix #1 has the greatest diversified VaR. Starting from Mix #1, risk is reduced by ADDING to the component with the lower marginal VaR (in this mix, the marginal VaR of the Tech ETF is only 0.020 while the marginal VaR of the REIT ETF is 0.461) and REDUCING the component with the higher marginal VaR. If we thusly start with Mix #1 and add 15.0% to the Tech ETF, then we “arrive” at Mix #2. However, in Mix #2 the Marginal VaR of the REIT ETF is still much higher than the Marginal VaR of the Tech ETF (i.e., 0.386 versus 0.118) so we can continue to shift toward the position with the lower Marginal VaR (which is still the Tech ETF). If we add another 15% to the Tech ETF, then we arrive at Mix #3: here the Marginal VaRs are almost equal, so we are very close to the allocation that minimizes the portfolio’ s diversified VaR. For the record, the diversified portfolio VaR for the four mixes are \$6,131.14 (Mix #1), \$5,043.03 (Mix #2), \$4,621.42 (Mix #3), and \$5,036.58 (Mix #4).

考察: Finance: Risk Management      重要程度: [■ ■ □]

科目: Quantitative Analysis

**31.** In comparison to a yield-based DV01 hedge, which of the following is the **best** argument in favor of a regression hedge?

- A. The regression hedge minimizes the profit and loss (P&L) variance of the hedged position
- B. The regression hedge is based on economic theory and future developments rather than empirical analysis
- C. The regression hedge does not require any out-of-sample test(s), which are often unreliable if the sample is small

D. The regression hedge solves for a risk-weight which qualifies as a valid risk-weighted asset (RWA) under Basel regulations

答案:A

解析:

**A 选项正确。** True: The regression hedge minimizes the profit and loss (P&L) variance of the hedged position. As Tuckman explains (emphasis ours), “The **best argument for the regression hedge** is actually not the earlier assumption that bond yields change exactly according to the regression model. Write the P&L of the position as ... formula 6.11 ... where the negative signs reflect that a positive face amount with a positive change (i.e., increase) in yield lowers P&L. It can then be shown that the regression hedge in Equation (6.8) minimizes the variance. In other words, **to the extent that P&L variance is an appropriate measure of risk, the regression hedge minimizes the risk of the hedged position.**” –Tuckman, Bruce; Serrat, Angel. Fixed Income Securities (Wiley Finance) (p. 159). Wiley. Kindle Edition. In regard to (B), (C), and (D), each is FALSE. Instead, the following are true statements:

- The regression hedge is based on empirical analysis but not economic theory and future developments
- The regression hedge typically does require (or at least prefer) out-of-sample test(s)
- The risk weight is the beta coefficient and has nothing to do with risk-weighted asset (RWA) under Basel regulations

考察: Fixed Income: Hedging      重要程度: [■ ■ □]

科目: Quantitative Analysis

**32.** Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among these (the following) four positions, which is the BEST example of wrong-way risk?

- A. Zapex has written (aka, sold) a call option to a counterparty in the same industry
- B. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery
- C. Zapex has written (aka, sold) a credit default swap (CDS) to counterparty whose credit quality is positively correlated with the reference entity
- D. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability

答案:D

解析:

**D 选项正确。** Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability Wrong-way (credit) risk is adverse correlation between the exposure and the credit quality, so wrong-way risk is a positive (negative) correlation between exposure and default probability (credit quality). This is wrong-way risk because greater exposure is associated with a lower likelihood of payment. Algebraically, wrong-way (or right-way) risk is some dependence between exposure/EAD and PD. We normally write  $EL\% = PD \times EAD$  but that assumes independence between PD and EAD, so a way to think about WWR is simply when  $EL > PD \times EAD$ , because  $EL \neq PD \times EAD$  is also a condition for non-independence. In regard to (A), (B) and (C), each is an example of RIGHT-WAY risk.

考察: Credit Risk: Wrong-Way Risk      重要程度: [■ ■ □]

科目: Quantitative Analysis

**33.** According to the FDIC, model risk management starts with robust model development, implementation, and use (aka, the first element. The second element is a sound validation process. The third element is governance). Which of the following is a TRUE statement about this *first element* (development, implementation and use) of model risk management?

- A. Model development should be a straightforward and routine technical process
- B. To exercise conservatism, pick an extreme point on a given modeled distribution

C. Banks should never model output with judgmental or qualitative adjustments

D. Although the nature of testing varies by the context and type of model, testing is an integral part of model development

答案:D

解析:

**D. TRUE:** Although the nature of testing varies by the context and type of model, testing is an integral part of model development

In regard to (A), (B) and (C), each is FALSE.

- Model development is neither straightforward nor a routine technical process!
- To pick an extreme point on a given modeled distribution is not necessarily conservative if the distribution is mis-specified or mis-estimated in the first place.
- Judgmental or qualitative adjustments are a KEY part of model development/implementation/use: “In addition, a considerable amount of subjective judgment is exercised at various stages of model development, implementation, use, and validation. It is important for decision makers to recognize that this subjectivity elevates the importance of sound and comprehensive model risk management processes.” ... “Banks should ensure that the development of the more judgmental and qualitative aspects of their models is also sound. In some cases, banks may take statistical output from a model and modify it with judgmental or qualitative adjustments as part of model development.”

考察: Operational Risk: Model Risk Management      重要程度: ☒☒☐

科目: Quantitative Analysis

**34.** Suppose that Acme Bank reports interest-sensitive assets (ISA) of \$490.0 million and interest-sensitive liabilities (ISL) of \$610.0 million. Recall that the relative interest-sensitive gap (aka, Relative IS GAP) is equal to the Dollar IS GAP divided (aka, scaled) by the size of the bank (where IS assets is a valid measure of size). Respectively, what is the bank’s relative IS GAP; its interest-sensitivity ratio (ISR); and the impact of rising interest rates on its net interest margin (NIM)?

- A. The Relative IS GAP is -120.0; the ISR is 1.245; rising interest rates will lower the NIM
- B. The Relative IS GAP is -0.245; the ISR is 0.803; rising interest rates will lower the NIM
- C. The Relative IS GAP is -0.245; the ISR is 1.245; rising interest rates will increase the NIM
- D. The Relative IS GAP is +4.08; the ISR is 0.803; rising interest rates will increase the NIM

答案:B

解析:

**B 选项正确。** The Dollar IS GAP = ISA - ISL = \$490.0 - \$610.0 = -120.0. The Relative IS GAP =  $-120.0/490.0 = -0.245$ . The ISR =  $490.0/610.0 = 0.803$ . This bank is liability-sensitive such that rising interest rates will lower its net interest margin (NIM).

考察: Market Risk: Interest Rate Risk      重要程度: ☒☒☐

科目: Quantitative Analysis

**35.** You work for the Private Client division of an international bank. The bank’s Private Client Advisors work with high net worth (NNW) individuals and families. Your boss has tasked you to perform due diligence on a roster of hedge funds; the funds that survive your due diligence will become candidates for presentation (and recommendation) to suitable customers of the bank who will become investors in the fund(s).

Your boss says that your due diligence should include a careful screen for fraud risk on behalf of the investors. In the case of your due diligence, which of the following is the most important ingredient?

- A. Diligent application of a comprehensive checklist including a list of specific questions
- B. Strong pre-existing personal relationships among the fund’s managers and all related parties

- C. Guarantees by the fund of exceptional performance (e.g., > 30% per annum) with little or no risk
- D. The investors’ membership in an affinity group, preferably a religious group due to high correlation with ethics

答案:A

解析:

**A. TRUE:** Diligent application of a comprehensive checklist including a list of specific questions  
The authors recommend reviewing the SEC website including its list of questions; e.g., Red Flags of Investment Fraud Checklist at <https://www.investor.gov/>.  
In regard to (B), (C) and (D), each is false:

- Strong pre-existing personal relationships with related parties is the **opposite of the independence** that investors should require between fund managers and related parties.
- Guarantees of exceptional performance with little or no risk is a classic **red flag**.
- In regard to affinity groups: these are notorious targets of fraud because participants tend to be trusting of their peers.

考察: Finance: Due Diligence      重要程度:

科目: Valuation and Risk Models

**36.** Which of the following is the BEST summary of the purpose of a copula function?

- A. To utilize the Gaussian correlation matrix such that tractable pairwise correlation parameters replace intractable dependence values
- B. To specify a single multivariate distribution as a function of various (even unusual) univariate marginal distributions by mapping them to distributions with an accessible or familiar dependence structure
- C. To translate a vector of outcomes into quantiles and map the quantiles to probabilities such that virtually any combination of multiple outcomes can be expressed as a single vector
- D. To decompose a posterior conditional distribution into a combination of the prior probability, the likelihood, and the marginal probability, typically by identifying Cholesky’s decomposition

答案:B

解析:

**B 选项正确。** To specify a single multivariate distribution as a function of various (even unusual) univariate marginal distributions by mapping them to distributions with an accessible or familiar dependence structure

考察: Quantitative Analysis: Copulas      重要程度: [■□□□□]

科目: Foundations of Risk Management

**37.** Retail exposures include the following: credit cards, installment loans (e.g., personal finance, educational loans, auto loans, leasing), revolving credits (e.g., overdrafts, home equity lines of credit), and residential mortgages. Each of the following tends to be a stronger feature of retail credit risk rather than corporate credit risk EXCEPT which is more typical of corporate credit risk?

- A. Portfolio tends to behave like a well-diversified portfolio in normal markets
- B. Default by a single customer is never expensive enough to threaten a bank
- C. A rise in defaults is often signaled in advance by a change in customer behavior
- D. Portfolio risk is dominated by risk that credit losses will rise to unexpected level due to concentrations of exposures

答案:D

解析:

**D 选项正确。** This tends to be a feature of CORPORATE credit risk: portfolio risk is dominated by risk that credit losses will rise to unexpected level due to concentrations of exposures.



- In regard to (A), (B) and (C), each is a tendency of retail credit risk.

考察: Credit Risk: Retail vs Corporate      重要程度: [■ ■ □]

科目: Quantitative Analysis

**38.** According to the Basel Committee on Banking Supervision (BCBS, 2018), in regard to cyber-security and cyber-resilience metrics, “Some jurisdictions have methodologies to assess or benchmark regulated institutions’ cyber-security and resilience ... None of these methodologies produce quantitative metrics or risk indicators comparable to those available for financial risks and resilience; e.g., standardized quantitative metrics where established data are available. Instead, indicators provide information on regulated institutions’ approach to building and ensuring cyber-security and resilience more broadly. Supervisory authorities also rely on entities’ own management information, although this differs across entities and is not yet mature.” (†)

With respect to current cybersecurity and resilience metrics, each of the following statements is true EXCEPT which is inaccurate?

- A. No single cybersecurity/resilience metric in isolation is sufficient
- B. Patch aging (aka, days to patch) is a widespread and comparable metric
- C. Backward-looking metrics are helpful but insufficient; forward-looking metrics are necessary because adversaries dynamically adapt
- D. The penetration ratio (i.e., percentage of banks in a jurisdiction that can be penetrated) is a useful metric that compares a supervisor’s effectiveness to other jurisdictions

答案:D

解析:

**D 选项正确。** False. The displayed penetration ratio is nonsensical (also to compare supervisor effectiveness), but penetration tests (penetration testing) are discussed several times by BCBS as an effective cyber-risk control approach. However, please do note that penetration testing (by type: count and finding rating) is a common and effective cyber-resilience metric. See 4.2.2. Penetration testing.

In regard to (A), (B) and (C), each is TRUE. Specifically:

- No single cybersecurity/resilience metric in isolation is sufficient: “Collectively, these indicators [i.e., cyber-metrics collated by regulated entities] can inform on the broad adequacy of an institution’ s cyber- and operational resilience levels for its business needs and risk appetite. However, no single item taken in isolation is seen as a sufficient metric, and no standard set of indicators has been identified so far to provide a meaningful benchmark.” (†)
- Patch aging (aka, days to patch) is a widespread and comparable metric: “Annex C contains cyber-centric metrics collated by a sample set of regulated institutions for decision-making bodies (boards and board sub-committees). It is notable that the data provided typically allow for trend information so that the reviewer can assess if the situation is getting better or worse. Some metrics track compliance with internal policies while others measure inherent risk. Patch ageing in particular is a widespread and comparable metric.” (†)
- Backward-looking metrics are helpful but insufficient; forward-looking metrics are necessary because adversaries dynamically adapt: “Backward-looking indicators comment on past performance as an indicator of future performance, which is reasonable when institutions’ operations and risk environment are relatively stable over time and more or less independent from outside influences. However, cyber-risk frustrates this because adversaries are dynamic, themselves adapting to institutions’ responses and protective measures, sometimes changing their tactics and strategies even in the space of a single cyber-incident. Distributed denial of service (DDOS) incidents are a good example, where the volume and scale of disrupted internet traffic generated has increased significantly in the last two years and adversaries adapt their techniques in response to an institution’ s defences. While backward-looking metrics continue to be important, jurisdictions are increasingly recognising the need for forward-looking indicators as direct and indirect metrics of resilience, indicating whether a regulated institution is likely to be more or less resilient in the event of a risk crystallising.” (†)

考察: Operational Risk      重要程度: [■ ■ □]



39. Venkat explains that “all risk management frameworks start with a governance structure that defines the roles and responsibilities of various bank employees and committees in overseeing risk-related activities,” and effective governance includes the oversight of intraday liquidity risk. In regard to the governance structure of intraday liquidity risk management which of the following statements is TRUE?
- A. Intraday liquidity risk should be accepted as a cost of doing business
  - B. Roles and responsibilities should be defined within the eight lines (i.e., two times four) of defense model
  - C. PCS Systems should only be a source of funds; if PCS becomes a use of funds, then a yellow flag should be triggered
  - D. Intraday liquidity risk should be incorporated in the risk taxonomy and is a component of risk self-assessment including settlement risks.

答案:D

解析:

**Option D is correct.** Intraday liquidity risk should be incorporated in the risk taxonomy and is a component of risk self-assessment including settlement risks.

In regard to (A), (B) and (C), each is FALSE.

- Venkat(†) explains that Active Risk Management is characteristic of the governance of intraday LRM: “Active risk management. In many institutions, intraday liquidity risk is accepted as a cost of doing business and is not as actively managed with the same level of rigor as other types of enterprise risk or even other liquidity risks. The leading banks with large volumes of PCS activities recognize the criticality of understanding and working to reduce their intraday liquidity risks. These institutions classify settlement and systemic risks as components of their risk taxonomy, and critically, incorporate them into the firm’ s risk appetite framework” (†)
- Integration with risk governance includes the THREE lines of defense model: Treasure, Corporate Risk Management, and Internal Audit. Says Venkat(†), “The intraday liquidity risk management framework follows the industry’ s three lines of defense model, with particular emphasis on expertise in the second line of defense to coordinate across the institution ... Corporate Risk Management is the second line of defense, responsible for overseeing funding-related policy and procedures: advising in the development of risk management programs, monitoring the ongoing risk-taking activities across all of a bank’ s funding desks, aggregating reporting across the bank, and providing an independent view of the effectiveness of the bank’ s overall intraday liquidity risk management programs.” (†)
- Payment, clearing, and settlement (PCS) Systems can be either a source or use of funds: “Settlements at PCS systems. Most PCS systems have one settlement per day, with many occurring in the late afternoon timeframe. This may serve as either a source or use of funds depending on the net position of a participant on any given day.” (†)

考察: Liquidity Risk Management      重要程度: [■■■]

科目: Valuation and Risk Models

40. Hedge funds are less regulated than mutual funds. According to the authors (Fung and Hsieh), this lighter regulation of hedge funds offers genuine benefits but also creates unintended consequences. Which of the following is both a negative consequence and a TRUE statement about this relatively light regulation?
- A. Lack of reliable, standardized, and long-term performance reporting
  - B. Lack of agreement on the specific definitions of absolute return and alpha
  - C. Ability of global macro funds to secretly drift into various strategies and styles with reckless impunity
  - D. Ability of hedge funds to advertise (and specifically solicit) to individual investors without any restrictions

答案:A

解析:

A 选项正确。 Lack of reliable, long-term performance reporting

According to the authors (emphasis ours): “However, being lightly regulated does have unintended consequences. For one, **performance records of hedge funds are generally not standardized and available reports are prone to measurement errors**. Until the early 1990s, the historical performance of hedge funds was often as private as their investors, and the assessment of hedge fund performance was as much art as science ... the **events of 1994** prompted investors to re-examine the perceived diversification of their hedge fund holdings. In turn, this aided the development of electronically available hedge fund databases with which more formal analysis of hedge fund performance and the attendant risks can be conducted. It was the arrival of electronic databases that made academic research in hedge funds a feasible proposition.” —Source: Chapter 17 of Handbook of the Economics of Finance, Volume 2B, edited by G. Constantinides, M. Harris, and R. Stulz.

- In regard to (B) and (D), these are false statements.
- In regard to (C), it is a feature (aka, advantage) of global macro funds that they can utilize various styles opportunistically.

考察: Finance: Hedge Funds      重要程度: [■■■]

科目: Quantitative Analysis

41. Meissner conducted a detailed study of equity correlations of the 30 stocks in the Dow (DJIA) over 44.5 years (i.e., the 534 months from January 1972 to July 2017) that generated  $534 \times 30^2 = 480,600$  correlations values. In regard to his empirical findings with respect to equity correlations, each of the following statements is TRUE, except which is false?

- A. Mean reversion should be included in equity correlation models
- B. For equity correlations, the Johnson SB distribution is better than the normal or lognormal
- C. A stressed scenario of bad economic times should incorporate higher correlation levels and higher correlation volatility
- D. Autocorrelation is a monotonically increasing function of the monthly lag; i.e., higher lags imply higher autocorrelations

答案:D

解析:

D 选项正确。 D. FALSE. Instead, the autocorrelation tends to decay (i.e., decrease) with greater lags. In regard to (A), (B), and (C), each is TRUE.

考察: Quantitative Analysis: Correlation      重要程度: [■■■]

科目: Quantitative Analysis

42. Your colleague has drafted guidance for your firm’s computation of the credit value adjustment (CVA). The guidance begins with four principles: risk-neutrality, discounting, proportionality, and inseparability. The principles are summarized as follows:

- I. Risk-neutrality: CVA can be computed either with risk-neutral (aka, market-implied) or real-world (aka, historical) parameters
- II. Discounting: CVA should be discounted to present value, either via discounted expected exposure (EE) or via explicit discount factors
- III. Proportionality: CVA will be proportional to the likelihood of default (PD), the exposure (EE) and the percentage amount lost in default (LGD).
- IV. Inseparability: It should be impossible to separate the valuation of a transaction from its counterparty risk as measured by CVA

Each of the principles is valid, EXCEPT which of the principles is INCORRECT?

- A. Risk-neutrality
- B. Discounting
- C. Proportionality

D. Inseparability

答案:D

解析:

**D 选项正确。** FALSE (Inseparability). Instead, the CVA is separable and this has practical importance. The most fundamental, key relationship is given by:  $\text{Risk value} = \text{Riskfree value} - \text{CVA}$ . In regard to (A), (B) and (C) each is TRUE as a principle.

考察: Counterparty Credit Risk      重要程度: ☒☒☐

科目: Quantitative Analysis

**43.** Acme Financial is a large bank whose Business Indicator (BI), for purposes of operational risk capital under the standardized measurement approach (SMA), is well in excess of EUR 1.0 billion. Therefore, Acme is required to use loss data as a direct input into its operational risk capital calculations. In regard to the general and/or specific criteria recommended by the Basel Committee, which of the following is TRUE about the identification, collection, or treatment of operational loss data?

- A. P&L provisions or reserves against potential loss impacts are included in the gross loss computation
- B. All internal loss events must be included in the gross loss computation regardless of the size of the loss
- C. Legal expenses are excluded from the gross loss computation in the operational loss data because they represent the realization of legal risk
- D. Receivables (a short-term asset) should be counted as recoveries unless the particular receivable causes the current ratio to exceed an internal limit

答案:A

解析:

**A 选项正确。** True: P&L provisions or reserves against potential loss impacts are included in the gross loss computation

In regard to (B), (C), and (D), each is FALSE. Rather, the following is true:

- While loss data must capture “all material activities,” losses below the minimum threshold do not need to be included.
- Legal expenses directly related to the event are included (see below). Recall that operational risk INCLUDES legal risk
- Receivables do not count as recoveries (see below).

From Gross loss, net loss, and recovery definitions (under 6. Specific criteria on loss data identification, collection and treatment):  
“21. Gross loss is a loss before recoveries of any type. Net loss is defined as the loss after taking into account the impact of recoveries. The recovery is an independent occurrence, related to the original loss event, separate in time, in which funds or inflows of economic benefits are received from a third party.

22. Banks must be able to identify the gross loss amounts, non-insurance recoveries, and insurance recoveries for all operational loss events. Banks should use losses net of recoveries (including insurance recoveries) in the loss dataset. However, recoveries can be used to reduce losses only after the bank receives payment. Receivables do not count as recoveries. Verification of payments received to net losses must be provided to supervisors upon request.

23. The following items must be included in the gross loss computation of the loss data set:

- (a) Direct charges, including impairments and settlements, to the bank’s P&L accounts and write downs due to the operational risk event;
- (b) Costs incurred as a consequence of the event including external expenses with a direct link to the operational risk event (eg legal expenses directly related to the event and fees paid to advisors, attorneys or suppliers) and costs of repair or replacement, incurred to restore the position that was prevailing before the operational risk event;
- (c) Provisions or reserves accounted for in the P&L against the potential operational loss impact;
- (d) Losses stemming from operational risk events with a definitive financial impact, which are temporarily booked in transitory and/or suspense accounts and are not yet reflected in the P&L ( “pending losses” ). Material pending losses should be included in the loss data set within a time period commensurate with the size and age of the pending item; and
- (e) Negative economic impacts booked in a financial accounting period, due to operational risk events impacting the cash flows or financial statements of previous financial accounting periods (timing losses” ). Material “timing losses” should be included

in the loss data set when they are due to operational risk events that span more than one financial accounting period and give rise to legal risk.”

(Source: Basel III: Finalizing post-crisis reforms, Basel Committee on Banking Supervision Publication (December 2017), page 132.)

考察: Operational Risk      重要程度: [■ ■ □]

科目: Quantitative Analysis

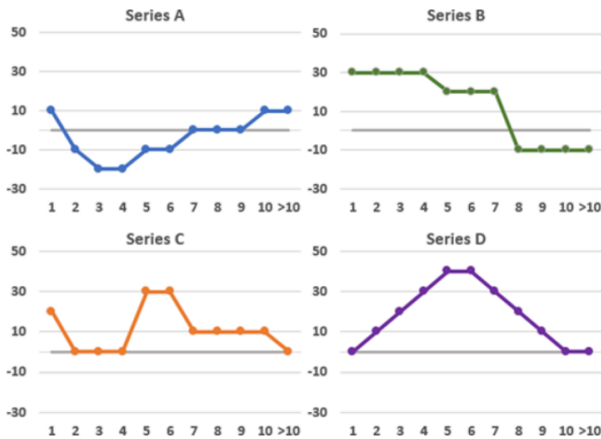
44. Geofinancial Bank currently has the following (very) simplified balance sheet:

| Assets |        | Liabilities |        |
|--------|--------|-------------|--------|
| Bonds  | \$30.0 | Deposits    | \$30.0 |
| Loans  | \$70.0 | Bonds       | \$50.0 |
|        |        | Equity      | \$20.0 |

Further, the maturities of these accounts are as follows:

- **Assets:** The bonds (\$30.0 million) expire in one year. In regard to the loans (\$70.0), \$40.0 million expire in five (5) years, \$10.0 million expire in seven (7) years, and \$20.0 million expire in ten (10) years.
- **Liabilities:** In regard to the deposits (\$30.0 million), \$10.0 million expire in one (1) year, and \$20.0 million expire in two (2) years. In regard to the bonds (\$50.0 million), \$10.0 million expire in five (5) years, \$30.0 million expire in seven (7) years, and \$10.0 million expire in beyond ten (> 10) years.
- **Equity** (\$20.0 million) is presumed to expire in ten (10) years

Consider the following possible term structures of expected cash flows:



Which term structure of expected cash flows is accurate for Geofinancial Bank?

- A. Series A
- B. Series B
- C. Series C
- D. Series D

答案:A

解析:

A 选项正确。 True: Series A. See below.

| Liabil-  |        |       |          |     |
|----------|--------|-------|----------|-----|
| Maturity | Assets | ities | $\Delta$ | Net |
| 1        | 30     | 10    | 20       | 20  |
| 2        |        | 20    | -20      | 0   |
| 5        | 40     | 10    | 30       | 30  |
| 7        | 10     | 30    | -20      | 10  |
| 10       | 20     | 20    | 0        | 10  |
| >10      |        | 10    | -10      | 0   |
| Sum      | 100    | 100   |          |     |

考察: Finance: Asset Liability Management      重要程度: [■■■□]  
 科目: Quantitative Analysis

45. Acme Industrial Corporation just finished establishing its new financial accounting controls which includes three components: planning, budgeting, and variance monitoring. The company’s next step is to establish a three-legged risk management stool that will analogously include the risk plan, the risk budget, and the risk monitoring process. In regard to the risk plan, each of the following will be an element EXCEPT which is the LEAST likely to be included in Acme’s risk plan?

- A. Diversification or risk decomposition policy
- B. Strategic responses for events that are low-probability but franchise-threatening
- C. Ample budget for losses that are material, predictable, and avoidable that cascade down
- D. Identification of critical dependencies that exist inside and outside the organization

答案:C  
 解析:

C 选项正确。 FALSE. The risk plan should have zero tolerance for mistakes/losses that are “material, predictable, and avoidable.”

In regard to (A), (B), and (D) each is TRUE. According to the authors, the five guideposts in the risk plan are:

- Expected return and volatility goals; and mile posts to recognize failure.
- Definition of success or failure
- Vision for how risk capital will be deployed; and a diversification or risk decomposition policy
- Bright line definition between “merely disappointing events” and seriously damaging losses. Further, “Strategic responses should exist for any franchise-threatening event—even if such events are low-probability situations. The risk plan should identify those types of losses that are so severe that insurance coverage (e.g., asset class puts) should be sought to cover the downside.” (source: 2020 Financial Risk Management Part II: Risk Management and Investment Management, 10th Edition. Pearson Learning Solutions, 2019. VitalBook file.)
- Identification of critical dependencies inside and external to the organization

考察: Financial Risk Management      重要程度: [■■■□]  
 科目: Quantitative Analysis

46. Bertha has a view about correlation, and she is evaluating several rainbow (aka, multi-asset) options. Her view is that correlation is going to decrease. Among the following trades, which **best** expresses her view?

- A. Buy (aka, long position in) basket option
- B. Buy (aka, long position in) better-of-two where  $\text{payoff} = \max(S_1, S_2)$
- C. Buy correlation (aka, pay the fixed rate) in a correlation swap
- D. Short position in an exchange option where  $\text{payoff} = \max(0, S_2 - S_1)$

答案:B

解析:

**B 选项正确。** Bertha wants to be “short correlation,” and the better-of-two options become more valuable as the correlation decreases. On the other hand:

- The basket option’s value becomes less valuable as the correlation decreases, such that buying the basket option is to be long correlation
- Buying correlation (aka, paying the fixed rate) in a correlation swap is long correlation
- An exchange option becomes more valuable as the correlation decreases such that a long position in an exchange option is short correlation, and short position in an exchange option is long correlation

考察: Market Risk: Correlation      重要程度: ☒☒☐

科目: Quantitative Analysis

**47.** Julia the analyst is analyzing the credit risk of a contract to which her firm is considering entering into as a party. If true, which feature of her work (among the choices below) **most likely** signifies or indicates that this is a case of counterparty credit risk rather than (traditional) lending risk or some other type of risk?

- A. Interest rate (as a market variable) fluctuations have little impact on the amount owed
- B. The exposure at default (EAD) is 100% of the notional over most of the life of the contract
- C. Her firm is the only party assuming credit risk: if her firm defaults, the other party faces no losses
- D. She is employing Monte Carlo simulation because the future value of the contract is highly uncertain

答案:D

解析:

**D 选项正确。** She is employing Monte Carlo simulation because the future value of the contract is highly uncertain.

According to Jon Gregory, the two key differences between counterparty and lending risk are (1) the lender knows the amount at risk but in the case of counterparty risk the amount is highly uncertain, and (2) lending involves unilateral risk (by the lender) but counterparty risk is bilateral to both counterparties.

In regard to false (a), Herstatt risk is a special case (cross-currency) of settlement risk as opposed to pre-settlement risk.

考察: Counterparty Credit Risk      重要程度: ☒☒☐

科目: Quantitative Analysis

**48.** The Basel regulatory framework has undergone several major renovations but started in the late 1980s when the Basel Committee on Banking Supervision published the first Basel accord, which is now called Basel I; this original Basel I was implemented in 1992. According to Carey (GARP Chapter 19), which BEST summarizes the motivation(s) of the original Basel?

- A. To formalize regulatory capital requirements with a set of regulations (i.e., Basel) that would become legal requirements and therefore carried the force of law; i.e., the motivation was legal
- B. Regulators realized that specific measures of short- and long-term liquidity were necessary given that liquidity crunches had been the essential common denominator of recent financial disasters; i.e., the motivation was liquidity

- C. The growth of cross-border finance required both the solvency of banks during stressful periods and fair competition (aka, level playing field) between banks competing in each other’s home countries; i.e., the motivations were bank safety and competitive concerns
- D. In the wake of a global recession, there was a motivation to incentivize banks to increase their lending (which had been historically low) in order to grow the developed and emerging economies; i.e., the motivation was economic recovery especially of emerging economies

答案:C

解析:

**Option C is correct.** True: The growth of cross-border finance required both the solvency of banks during stressful periods and fair competition (aka, level playing field) between banks competing in each other’s home countries; i.e., the motivations were bank safety and competitive concerns.

Carey explains, “Two events motivated creation of Basel I.

- First, the growth of cross-border finance continued after Herstatt’s failure and it was evident that the G10 nations had a common interest in ensuring that banks had enough equity to absorb large losses.
- Second, international banks were competing vigorously in each other’s home countries. However, minimum levels of required capital varied significantly across nations, creating a perception that banks headquartered in countries with low minimums had a competitive advantage. In response, members of the BCBS decided to develop a global minimum standard to “level the playing field” and avoid a race to the bottom. That is, while the Basel Accord was partly about ensuring safety and soundness, negotiations also had an element of maneuvering for perceived competitive advantage.”(Source: Chapter 19, 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition. Pearson Learning Solutions.)

考察: Finance: Basel Accords      重要程度: [■■■□]

科目: Quantitative Analysis

**49.** Which of the following types of liquidity is available in the form of the institution’ s liquidity asset buffer (LAB) and represents liquidity available to meet general financial obligations under a stress scenario?

- A. Operational
- B. Strategic
- C. Contingent
- D. Restricted

答案:C

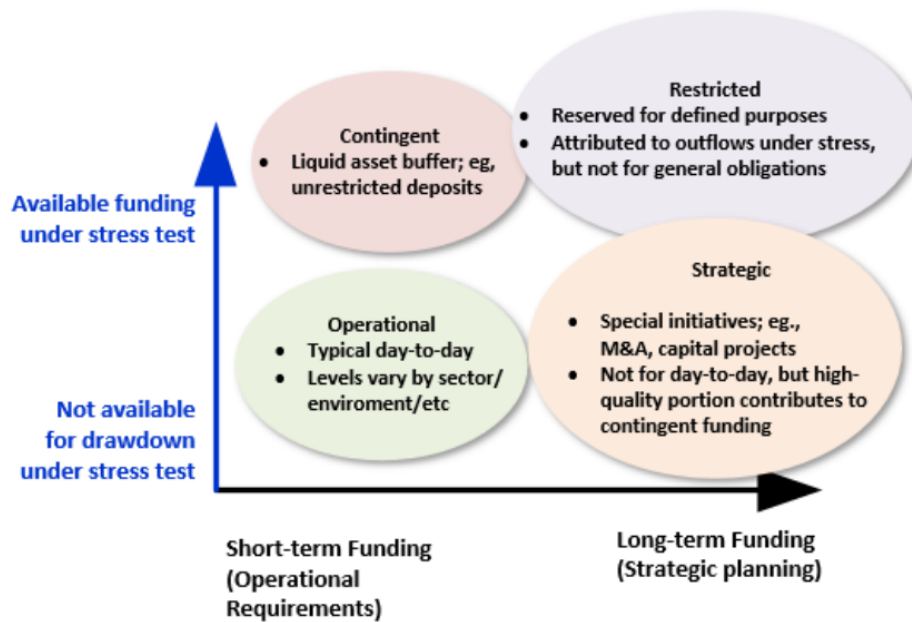
解析:

**C 选项正确。** Contingent liquidity captures cash or assets that are available when a stressing scenario materializes. The liquidity is contingent on the occurrence of the stress situation. Included within this category are financial assets that can be converted into cash without any material loss in value. Capturing contingent liquidity to cover stressed cash outflows represents the main reason for liquidity stress tests.

See also the diagram below (inspired by Venkat’ s Figure 3.1(†) but rendered by David Harper in MS Visio).

Also, writes Venkat (emphasis ours), “**Liquid asset buffer:** The liquid asset buffer represents the contingent liquidity that is currently in place ... The fundamental characteristics of liquid asset buffer securities should include low credit and market risk, ease and certainty of valuation, trading in an active and sizable market, and low concentration of buyers and sellers. The liquid asset buffer should also meet operational requirements that ensure the liquidity is under the control of the central treasury area of the entity undergoing the stress test.” (†)





考察: Finance: Liquidity Risk    重要程度: [■□□]  
 科目: Quantitative Analysis

50. Acme Investment Bank recently finished building the third leg of its three-legged risk management stool, after the risk plan and the risk budget: the bank installed its risk monitoring process. The bank shares a belief with the authors (Rosengarten and Zangari) that “risk is the shadow cost behind returns.” Each of the following is probably an element or plank in the bank’s risk monitoring process EXCEPT which is the least likely to be found in the bank’s risk monitoring process?

- A. The bank’s monitoring is not dependent on any single tool or metric, but instead relies on an ensemble of various tools and metrics
- B. The bank measures alpha as a regression intercept where alpha can help evaluate manager skill but unfortunately, its statistical significance often requires more data (i.e., longer history) than is available
- C. The bank measures return on risk capital (RORC)) where value at risk (VaR) is a measure of risk capital; in addition to the RORC’s level, the bank watches its variability
- D. After a threshold level of data is collected, subjective articulations of each portfolio manager’s philosophy becomes unnecessary, and should be discarded, in favor of quantitative performance evaluation

答案:D  
 解析:

**Option D is correct.** D. FALSE. Instead, articulation of each portfolio manager’s philosophy is always necessary as a complement to quantitative tools and to clarify whether style drift occurs.

As GARP explains in the reading, explicit (and subjective) articulation of philosophy is always a necessary complement to quantitative tools if for no other reason than it helps to identify *style drift*. From the reading: “Before discussing these [i.e., commonly used performance tools], however, it is worth noting that performance tools, while necessary, are not a substitute for timely management intervention when there is an indication of abnormal behavior. By the time that abnormal behavior manifests itself in the form of poor performance statistics, the damage might already be irreversible. For this reason, we believe that performance tools must be supplemented with:

- A clear articulation of management philosophy from each portfolio manager. This philosophy statement should identify how the manager expects to extract returns from the market. It should identify ways of knowing when the manager’s process is successful and when it is unsuccessful.
- A routine position and style monitoring process designed to identify deviations from philosophy or process. This is a type of early warning system.” —Source: 2020 Financial Risk Management Part II: Risk Management and Investment Management, 10th Edition. Pearson Learning Solutions, 2019. VitalBook file.

In regard to (A), (B), and (C), each is TRUE. Specifically,

- The bank’ s monitoring is not dependent on any single tool or metric, but instead relies on an ensemble of various tools and metrics
- The bank measures alpha as a regression intercept where alpha can help evaluate manager skill but unfortunately, its statistical significance often requires more data (i.e., longer history) than is available
- The bank measures return on risk capital (RORC)) where value at risk (VaR) is a measure of risk capital; in addition to the RORC’ s level, the bank watches its variability

考察: Financial Risk Management      重要程度: [■ ■ □]

科目: Quantitative Analysis

**51.** Acme FinGlobal Bank has engaged consultants to redefine the organization’ s risk measures. The project includes value at risk (VaR) mapping across several portfolios (aka, asset classes) at the organization:

- A fund that invests in initial public offerings (IPOs)
- A fund that invests in emergent assets but where with there is little historical data
- Small equity portfolio with less than 30 positions
- Large equity portfolio with 1,000 positions
- A portfolio of corporate bonds that carries six different ratings and has maturities of 3, 5, and 7 years

Each of the following statements is TRUE except which is false?

- A. The corporate bond can be mapped to nine general risk factors
- B. The IPO and emergent asset funds cannot be mapped because these instruments lack historical data such that sensitivities cannot be estimated
- C. The large equity portfolio is an ideal candidate for mapping because mapping to general and specific market primitives requires far fewer numerical estimates than specification of a covariance matrix
- D. The equity portfolio VaR mapping process contains sources of model risk, including factor sensitivities (aka, betas) estimation error, subjective risk factor choices, and specific risk omission

答案:B

解析:

**Option B is correct.** False. Instead, mapping is specifically a solution to the problem of insufficient data. As Jorion explains, “The risk manager would have to replace these positions by exposures on similar risk factors already in the system.” <sup>(†)</sup>

In regard to (A), (C), and (D), each is TRUE.

(†) Philippe Jorion, Value-at-Risk: The New Benchmark for Managing Financial Risk, 3rd Edition. (New York: McGraw Hill, 2007)

考察: Market Risk Measurement and Management      重要程度: [■ ■ □]

科目: Quantitative Analysis

**52.** Freecone Financial LLC entered into an interest rate swap some time ago where it agreed to make semi-annual fixed payments at a rate of 3.0% per annum, in exchange for receiving floating payments on a notional amount of \$50.0 million. The floating rate is the six-month secured overnight financing rate (SOFR) and this is also the discount rate used to value the swap. The swap has a remaining life of 1.25 years. The SOFR rate applicable to the exchange in three months was determined three months ago: it was 2.40%. Just as with a LIBOR-based swap, the floating rate is observed at the beginning of each six-month period but paid at the end of the period. Currently, the SOFR term structure is flat at 2.60% per annum with continuous compounding. With respect to its position in this swap, which of the following is nearest to Freecone Financial’ s current exposure?

- A. Zero
- B. -\$335,900
- C. +\$335,900
- D. +1,335,900

答案:A

解析:

**Option A is correct.** Current exposure =  $\max(\text{MtM value}, 0)$ . In this case,  $\text{CE} = \max(-335,900, 0) = 0$ . Freecone is underwater in the swap; its counterparty has positive current exposure because the counterparty experiences a loss in the event of default. But Freecone does not experience a CURRENT loss in the event of default.

考察: Credit Risk: Current Exposure      重要程度: [■■■□]

科目: Quantitative Analysis

**53.** According to the Guidance on Managing Outsourcing Risk (by the Board of Governors of the Federal Reserve System), with respect to a third-party service provider arrangement, the overall due diligence process by a financial institution should at least include a review of each of the following EXCEPT which is not essential?

- A. Ensure service provider has an appropriate background check program for its employees
- B. Financial review of service provider's most recent annual report and financial statements
- C. Evaluate the service provider's performance along environmental, social, and governance (ESG) factors
- D. Evaluate the adequacy of standards, policies and procedures including adherence to applicable laws, regulations and supervisory guidance

答案:C

解析:

**C 选项正确。** A review of ESG factors is not mentioned. According to the Board, the three broad review areas are:

1. *Business background, reputation, and strategy*: "Financial institutions should review a prospective service provider's status in the industry and corporate history and qualifications; review the background and reputation of the service provider and its principals; and ensure that the service provider has an appropriate background check program for its employees."
2. *Financial performance and condition*: "Financial institutions should review the financial condition of the service provider and its closely-related affiliates. The financial review may include: The service provider's most recent financial statements and annual report with regard to outstanding commitments, capital strength, liquidity and operating results; The service provider's sustainability, including factors such as the length of time that the service provider has been in business and the service provider's growth of market share for a given service ..."
3. *Operations and internal controls*.

考察: Operational Risk: Outsourcing Risk      重要程度: [■■■□]

科目: Quantitative Analysis

**54.** The staff at Umbrella Street Bank, which is a commercial bank (it could also be a savings institution or credit union), produces several liquidity risk reports on a daily, weekly, monthly, and quarterly basis. Among these liquidity risk reports is a deposit tracker report. Among the following metrics, which is MOST LIKELY to appear in their deposit tracker report?

- A. Loan-to-deposit (LTD) ratio (current and forecast) versus board-approved lower floor (limit) of 120.0%
- B. Loan-to-deposit (LTD) ratio (current and forecast) versus board-approved upper ceiling (limit) of 80.0%
- C. Market-to-book ratio of common equity (current and forecast) versus investor-communicated target of 1.30

D. The leverage-adjusted duration gap (current and forecast) versus board-approved upper ceiling of 3.5 years

答案:B

解析:

**B 选项正确。** True: Loan-to-deposit (LTD) ratio (current and forecast) versus board-approved upper ceiling (limit) of 80.0%  
In regard to (A), (C) and (D), each is FALSE.

考察: Liquidity Risk Management      重要程度: [■■■□]

科目: Quantitative Analysis

**55.** You work as a pension fund administrator and the Investment Committee asks for your input. The pension fund seeks to increase its allocation to alternative investments and, in particular, will add a new hedge fund strategy; aka, style. You are limited to four strategies. You are perusing the brochures of four hedge funds (each named after Greek gods per a popular marketing convention!). The four strategies are global macro, merger arb, long/short equity, and equity market neutral. Below are the four funds. Each fund has a name, a strategy (aka, style), a tagline, and a very brief description.

- *Acheilus* is a Global Macro strategy. We are your effective asset (factor) allocator! We opportunistically make bets in different markets utilizing a range of strategies; you can think of us as betting on a range of factors such as commodities, interest rates, exchange rates.
- *Boreas* is a Merger Arb (aka, Risk Arb) strategy. We collect nickels (and dimes) in front of the proverbial steamroller! Like our sibling event-driven strategy (i.e., Distressed), we can offer you nonlinear returns but with tail exposure to extreme market conditions.
- *Chimera* is a Long/Short Equity strategy. We are friendly but will never hug a benchmark! Although we have some directional market exposure, we can be classified as relative value because we are stock pickers who seek alpha by way of a diversity of opinions and ability.
- *Daphne* is an Equity Market Neutral strategy. We bring you a basket of betas! As an event-driven strategy, our returns are highly linked to several common market risk factors including equity indices, bond spreads, and volatility factors.

Each of these descriptions is consistent with its strategy (aka, style) category EXCEPT for one. Select the fund that is NOT consistent with its description.

- A. Acheilus
- B. Boreas
- C. Chimera
- D. Daphne

答案:D

解析:

**D 选项正确。** The tag/description is thrice inconsistent: Equity market neutral is unlinked to observable market risk factors (in the cited research, it is the ONLY strategy found without links to market risk factors). Consequently, an equity market-neutral fund would not “bring a basket of betas:” its intention is to neutralize betas; specifically, zero beta with respect to the common market factor. Further, it is not event-driven; equity market-neutral is non-directional and classified as a “niche strategy” by the authors, but can also be classified as “equity hedge.”  
In regard to (A), (B), and (C), each of the other funds (Acheilus, Boreas, and Chimera) have consistent strategies, taglines, and descriptions; i.e., they are conceivably true.

考察: Finance: Hedge Fund Strategies      重要程度: [■■■□]

科目: Quantitative Analysis

**56.** Betty’s firm wants to allocate an economic capital cushion based on a one-day 99.5% value at risk (VaR) model. Further, she will conduct a basic backtest (aka, exception or frequency test) where the frequency of tail losses fit a binomial distribution.

Her backtest window is limited to two years,  $T = 500$  days. In regard to this backtest, which of the following statements is TRUE?

- A. To increase the power of her backtest, she can decrease the VaR confidence level and apply a higher (k) capital multiplier
- B. At this high VaR confidence level and relatively short backtest window, the normal approximation is a better fit to test the model’s bias than the binomial distribution
- C. She can achieve a more powerful tests of the same hypothesis, and without the need for any additional information, by utilizing any of several extensions approved by regulators
- D. Although VaR is always a one-tailed quantile (e.g., the 95.0% standard normal VaR is always 1.645 and never 1.960), the backtest is always a two-tailed (2T) backtest such that Betty must have a lower and an upper cutoff

答案:A

解析:

**A 选项正确。** To increase the power of her backtest, she can decrease the VaR confidence level and apply a higher (k) capital multiplier.

A key statistical challenge for the VaR backtest is that a high confidence level implies a small sample of exceptions. A remedy is to lower the confidence level, but apply a multiplier (to the lower-confidence VaR) in order to approximate a higher-confidence VaR.

As Jorion explains, “The lack of power of this framework is due to the choice of the high VAR confidence level (99 percent) that generates too few exceptions for a reliable test. Consider instead the effect of a 95 percent VAR confidence level. (To ensure that the amount of capital is not affected, we could use a larger multiplier k.)” —Philippe Jorion. Value at Risk, 3rd Ed.: The New Benchmark for Managing Financial Risk (p. 150). Kindle Edition.

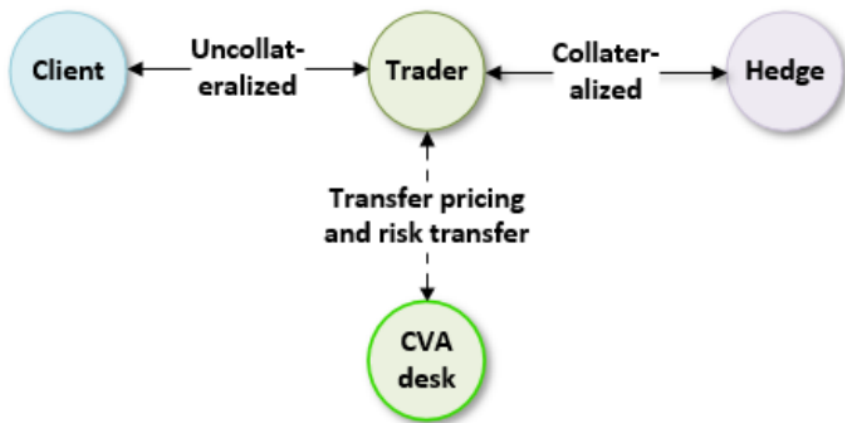
In regard to (B), (C) and (D), each is FALSE. Specifically,

- A general guideline for the normal approximation of the binomial is that it can be used when  $n \times p$  and  $n \times (1 - p)$  are greater than 10; some would say greater than 5. In this case,  $0.5\% \times 500 = 2.5$ , such that the binomial probably has too much skew.
- The readings (Jorion, Hull or even Dowd) do not reference power gains via regulator-approved extensions
- VaR is always a one-tailed quantile (e.g., the 95.0% standard normal VaR is always 1.645 and never 1.960), but the backtest is a statistical test and can be either one- or two-tailed (2T)

考察: Value at Risk (VaR)      重要程度: [■ ■ □]

科目: Quantitative Analysis

**57.** The figure below illustrates the general role of the credit value adjustment (CVA) desk at Heroic Multifunction Bank (aka, the bank). Assume the trader is a market maker. The uncollateralized trade with the customer (on the left) is hedged with a collateralized trade (on the right):



Assume the uncollateralized transaction (with the client) has a positive mark-to-market (MTM) value such that, by definition, the collateralized hedge position, therefore, has a negative MTM value. If the market moves such that, with respect to the position with the client, the position's positive MTM value INCREASES and becomes even more positive, then which of the following implications is most likely to be TRUE?

- A. Funding cost (FVA) is positive
- B. From the trader's (market maker's) perspective, CVA decreases
- C. Because the position is hedged, there is no associated regulatory cost of capital to the bank
- D. The trader (market maker) writes a contingent credit default swap (CCDS) to transfer its counterparty risk

答案:A

解析:

A 选项正确。 A. TRUE: Funding cost (FVA) is positive. It will be necessary to post collateral on the hedge and this amount will need to be funded. In regard to (B), (C) and (D), each is FALSE.

- In regard to false (B), CVA is likely to increase as exposure to the further positive position increases
- In regard to false (C), their hedge will not eliminate the need for regulatory capital! Says Gregory, “Capital: There will be capital requirements and these will change –which is important because capital is a cost. The capital for the uncollateralised transaction will increase as it becomes in-the-money. The capital for the hedge will not offset this due to being collateralised –and even if it was uncollateralised, there would likely be a convexity impact (the capital for one transaction would increase by more than the capital reduction on the other).”
- In regard to false (D), the problem here is that the trader needs to buy (purchase) not write (sell) a CCDS. Says Gregory: “More tailored credit derivative products such as contingent CDS (CCDS) have been designed to hedge counterparty risk even more directly. CCDSs are essentially CDSs but with the notional of protection indexed to the exposure on a contractually specified derivative (or even portfolio of derivatives). They allow the synthetic transfer of counterparty risk linked to a specific trade and counterparty to a third party. However, the CCDS market has never developed any significant liquidity.”

(Source: Jon Gregory, The xVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, 3rd edition (West Sussex, UK: John Wiley & Sons, 2015))

考察: Finance: Counterparty Credit Risk      重要程度: [■ ■ □]

科目: Quantitative Analysis

58. Andrew Ang makes an important, provocative statement when he writes “Reported illiquid asset returns are not returns.” He claims that people overstate the expected returns and understate the risk of illiquid assets, and he attributes this to three key biases. According to Ang, each of the following is a bias that overstates the expected returns (and/or understates the risk) of illiquid assets EXCEPT which is not accurate?

- A. Survivorship bias can inflate returns by 4.0% or more
- B. Infrequent sampling (aka, infrequent trading) artificially reduces risk and risk-related metrics such as volatility, correlation and beta
- C. Turnover bias decreases the typical time between transactions and tends to artificially increase the expected return by 5.0% or more
- D. Selection bias (aka, reporting bias) is a distortion of the sample that artificially increases (ie, overestimates) alpha and artificially decreases (ie, underestimates) beta

答案:C

解析:

**C 选项正确。** False. Turnover bias is made-up. In regard to (A), (B) and (D), each is TRUE. According to Ang , with respect to illiquid assets, the three biases that prompt investor to overstate expected returns while underestimating risk are: survivorship bias; infrequent sampling; and selection bias.

**考察:** Alternative Investments: Illiquid Assets      **重要程度:** [■■■□]

**科目:** Quantitative Analysis

**59.** Which of the following provides the BEST explicit link(s) between a bank’s contingency funding plan (CFP) and its liquidity stress testing framework?

- A. Communication plan
- B. Limit structure and escalation levels
- C. Funding sources concentration and non-core funding
- D. Industry factors such as profitability trends in the financial sector

**答案:**B

**解析:**

**B 选项正确。** True: limit structure and escalation levels

Explain Lai and Tuosto (emphasis ours), “II. Integrate with Broader Risk Management Frameworks: The CFP is not a stand-alone tool, but rather, an integrated part of the institution’s liquidity risk management and firm-wide risk management frameworks ... The CFP should be explicitly linked to the liquidity risk measurement framework and the liquidity stress test, in particular through its limit structure and escalation levels. For example, the liquidity risk measures used in the institution’s BAU risk management activities serve as a foundation from which the CFP defines its early warning indicators (EWIs). Additionally, linkages to the business continuity and crisis management frameworks will reinforce key operational and communication protocols during times of crisis.” —Chi Lai and Richard Tuosto, Contingency Funding Planning in Venkat’s Liquidity Risk Management

**考察:** Liquidity Risk Management      **重要程度:** [■■■□]

**科目:** Quantitative Analysis

**60.** Bodie, Kane and Marcus explain, based on an analogy originally developed by Robert C. Merton, that we can attempt to quantify the value of market timing by recognizing that foresight is similar to holding a call option on the owned equity portfolio (without paying the premium): the perfect market timer exercises her option, so to speak, by switching from the riskfree asset to the equity portfolio (or vice-versa) depending on which offers the better return. Consequently, the Black-Scholes (BSM) formula can be used, but when the strike price is given by  $\$1.00 \cdot \exp(rT)$ , the BSM call option formula conveniently reduces to  $c = 2N[0.5\sigma(e)\sqrt{T}] - 1$ .

If we assume annual forecasts ( $T = 1.0$  year), and the volatility of the equity portfolio,  $\sigma(e) = 36.0\%$  per annum, which of the following is **nearest** to the value of perfect market timing, as a percentage of the equity portfolio’s value?

- A. 7.39%
- B. 14.28%
- C. 46.50%
- D. 83.77%

**答案:**B

**解析:**

**B 选项正确。** In this case,  $0.5\sigma(e)\sqrt{T} = 0.5 \times 36.0\% \times \sqrt{1} = 18.0\% = 0.180$  and  $N(0.18) = 0.57140$  if rounded per the lookup table. Then  $c = 2 \times 0.57140 - 1 = 14.28\%$ .

**考察:** Portfolio Management      **重要程度:** [■■■□]

**科目:** Quantitative Analysis



**61.** Patricia has been asked by the Chief Risk Officer (CRO) to estimate value at risk (VaR) at high confidence levels. Initially, she fits a generalized extreme value (GEV) distribution to the loss data. However, some members of the ultimate audience (the board) do not understand the math. She observes that Dowd(†) explains, "[t]here are also short-cut ways to estimate VaR (or ES) using EV theory. These are based on the idea that if  $\xi > 0$ , the tail of an extreme loss distribution follows a power-law times a slowly varying function:  $F(x) = k(x)x^{-1/\xi}$  where  $k(x)$  varies slowly with  $x$ . For example, if we assume for convenience that  $k(x)$  is approximately constant, then [this equation] becomes:  $F(x) \approx kx^{-1/\xi}$ ". Notice the assumption that  $k(x)$  is approximately constant.

The exponent in Dowd’s power law maintains consistency with the generalized extreme value (GEV) distribution by referring to the exponent as a negative reciprocal,  $-1/\xi$ , but given this exponent is also a constant along with  $k(x)$ , we can further simplify the approximation: instead of  $F(x) \approx kx^{-1/\xi}$ , we can say that  $F(x) \approx kx^{-\xi}$  which is sometimes represented as  $F(x) \approx kx^{-\alpha}$  where  $\alpha = 1/\xi$ . For example, Patricia assumes the exponent’s value is -4.0. In the former syntax,  $\xi = 0.25$  such that  $(-1/\xi) = -4.0$ ; in the latter version,  $\alpha = 4.0$  such that  $(-\alpha) = -4.0$ , to the same effect. For our purpose here, we will simply refer to  $F(x) \approx kx^{-\xi}$ .

Consequently, she decides to estimate extremely confident VaRs with a power-law shortcut. Specifically, when Patricia sets xi,  $\xi = 4.0$ , she observes that the 99.50% VaR is \$6.0 million. With respect to the exponent, she assumes  $F(x) \approx kx^{-\xi} = F(x) \approx kx^{-4.0}$ . If the loss tail follows the power law, which is **nearest** to the implied 99.90% VaR?

- A. \$5.0 million
- B. \$9.0 million
- C. \$13.0 million
- D. \$21.0 million

答案:B

解析:

**B 选项正确。** We first need to solve for  $k$ . Dowd(†) uses an exponent of  $(-1/\xi)$  but we shall refer to its simpler reciprocal  $(-\xi)$  in the exponent. In this way, the power law says the following:  $\text{Prob}(v > x) = F(x) \approx kx^{-\xi}$ . Solving for the constant  $k$ , it must be that:  $k = F(x)x^{\xi}$ . In this case where we are given that  $x = 6.0$  when  $F(x) = 0.50\%$  and  $\xi = 4.0$ , we can find  $k = 0.50\% \cdot 6.0^4 = 6.480$ . Now we know, because we are assuming constant ( $k$ ) per the approximation, that our distribution is  $F(x) \approx kx^{-\xi} = 6.480x^{-4.0}$ . We want the VaR implied by  $F(x) = 0.10\%$  such that  $0.10\% = 6.480x^{-4.0}$  where  $x = (6.480/0.10\%)^{1/4.0} = 8.97$  which rounds to \$9.0 million.

考察: Quantitative Analysis: Extreme Value Theory      重要程度: ■■■□

科目: Quantitative Analysis

**62.** Acme Financial has a position in five derivative contracts with its counterparty. The current mark-to-market values of the positions are the following: +12, -9, +20, -6, and +8. The five positions are all included in a single netting set. Let’ s define the “netting benefit” as the reduction in current exposure due to netting; that is, the difference between the total current exposure with netting and the current exposure “as if” netting did not apply. What is the netting benefit in this set?

- A. Zero
- B. 3
- C. 15
- D. 25

答案:C

解析:

**C 选项正确。** True: 15. Without netting, total exposure =  $\max(0, +12) + \max(0, -9) + \max(0, 20) + \max(0, -6) + \max(0, +8) = 40$ . With netting, total exposure =  $+12 - 9 + 20 - 6 + 8 = 25$ . Notice this is the same as  $-(-9 - 6)$ .

考察: Counterparty Credit Risk      重要程度: ■■■□

科目: Quantitative Analysis

**63.** Patricia is a risk analyst who works for a large international bank. Her team is developing a new risk management framework for the bank and is currently working on risk identification. As part of extreme risk identification, Patricia has been assigned the important task of designing and performing the firm’s scenario analysis. Soon she will facilitate a brainstorming session for which she has prepared the following documents: external loss data, internal loss data (including near misses), risk and control self-assessment (RCSA) results, key risk indicator scores, audit issues, and concentrated exposures. Assuming she continues to follow best (or good) practice with respect to scenario analysis, which of the following is a TRUE statement?

- A. Because most large operational losses result from external causes, her primary focus should be scenarios driven by external causes
- B. She should compare her firm’s generated scenarios to industry-specific list(s) published by external sources, but such a comparison should be made after her firm’s own internal generation exercise
- C. The initial output of scenario identification should be limited to approximately 10 or 12 scenarios in order to avoid the fallacy of missing the forest for the trees
- D. Because the ultimate goal of scenario analysis is to predict the future with a sufficient degree of certainty, her process should acknowledge and elevate the role of human subjectivity

答案:B

解析:

**B 选项正确。** B. True: She should compare her firm’ s generated scenarios to industry-specific list(s) published by external sources, but such a comparison should be made after her firm’ s own internal generation exercise.According to GARP (section 3.2), “Finally, it can be useful for a firm to compare the scenarios it has generated with an industry list of scenarios, to determine whether it has omitted any relevant scenarios or risk drivers. The Operational Risk Consortium for Insurance Companies (ORIC) and ORX are examples of industry-specific organizations that provide ready-made scenario lists to their members. It is recommended, however, for an organization to compare its scenario list to a list of scenarios from an external industry source only after the scenario generation exercise and not before, so that the externally procured scenarios do not bias the generation process.” (†)

In regard to (A), (C), and (D), each is FALSE. Instead:

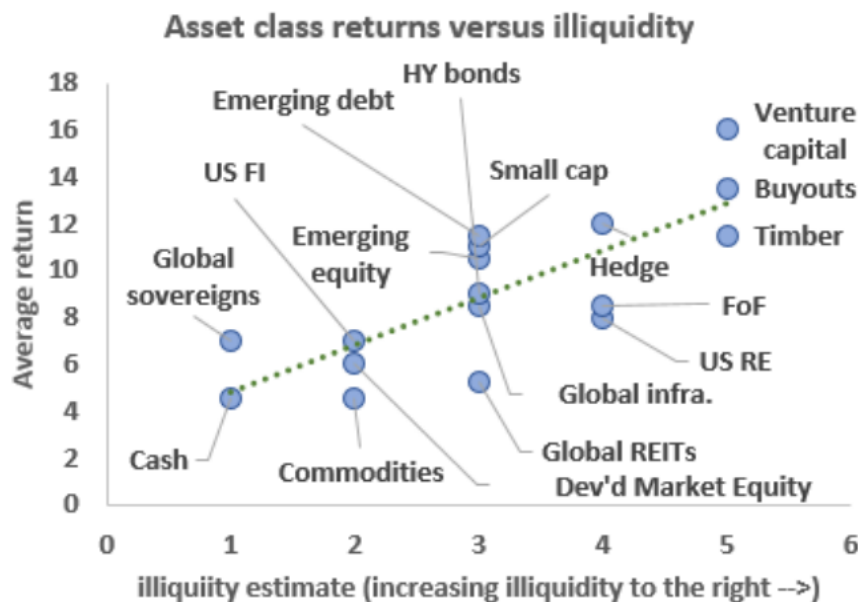
- Most large operational losses result from internal causes
- GARP recommends more scenarios, e.g., at least 20 to 30 scenarios for mid-sized firms. For large international banks, ~50 scenarios are not uncommon
- Scenario analysis is not about predicting the future but identifying a range of scenarios. Further, says GARP, “regulators demand that firms minimize subjectivity and biases as much as possible when conducting scenario identification and assessment.” (†)

(†) 2023 GARP FRM Part II: Operational Risk and Resilience, Pearson Learning Solutions

考察: Operational Risk     重要程度: [■ ■ □]

科目: Quantitative Analysis

**64.** In order to identify the presence of illiquidity risk premium(s), Andrew Ang(†) references data presented by Antti Ilmanen in his well-regarded book Expected Returns (An Investor’ s Guide to Harvesting Market Rewards). This data is displayed below as a scatterplot where the y-axis is the long-run average return of the asset class and the x-axis is an index of illiquidity. A higher index (ie, to the right) implies less liquidity. For example, the venture capital as an asset class is assigned to the least liquid (most illiquid) asset class but it also plots the highest long-run average return.



In regard to the illiquidity risk premium, which of the following statements is TRUE?

- A. In general illiquid asset classes offer high risk-adjusted returns
- B. These charts demonstrate that there do exist large illiquidity risk premiums ACROSS asset classes
- C. There do exist large illiquidity risk premiums WITHIN many asset classes, but not ACROSS asset classes
- D. Illiquid equities earn the same returns as liquid equities; and illiquid bonds earn the same returns as liquid bonds

答案:C

解析:

C 选项正确。 True: There do exist large illiquidity risk premiums WITHIN many asset classes, but not ACROSS asset classes.

In regard to (A), (B) and (D), each is FALSE.

考察: Finance: Portfolio Management      重要程度: [■ ■ □]

科目: Quantitative Analysis

**65.** Sally represents an investor who likes the strategy and style of a new, entrepreneurial start-up hedge fund. At her client's request, she is going to conduct due diligence. According to Kevin Mirabile (author of the hedge fund due diligence reading), as Sally conducts her diligence of the hedge fund on behalf of the investor, which of the following is the most important principle or ingredient?

- A. She should observe strong evidence that the fund has eliminated business model risk via the SEC's Zeta Sigma AIML filter
- B. She should confirm that the CEO is exceptionally good at all three top-level skills sets: investment, operational, and business
- C. She should follow all the steps in the process but with flexibility as due diligence is also an art and each fund is unique with its own strategy
- D. She should clarify that the fund's incentives are actual equity ownership grants rather than phantom equity and that they are granted throughout the organization to all full-time employees.

答案:C

解析:

C 选项正确。 She should follow all the steps in the process but with flexibility as due diligence is also an art and each fund is unique with its own strategy.

GARP explains that "performing due diligence on a hedge fund is as much art as it is science. An investor must use a comprehensive checklist to ensure that nothing is left out or omitted yet remain free to ask open-ended questions that provide insights into a firm's philosophy or culture. The guidelines discussed here for investment, operational, and business model risk assessment were for the most part generic. When performing due diligence, investors must modify their approach to be

strategy specific.” (Summary, 2020 Financial Risk Management Part II: Risk Management and Investment Management, 10th Edition. Pearson Learning Solutions, 2019. VitalBook file.)

In regard to false (A), (B) and (D):

- The Zeta Sigma AIML filter is fictional and business model risk cannot be eliminated
- The authors point out that only “dinosaurs” (as founders) attempt to manage all three because managers are rarely good at all three (nevermind exceptionally good)! Managers should not choose to be both CEO and CIO.
- Incentive structures vary by organization.

考察: Hedge Funds: Due Diligence      重要程度: [■ ■ □]

科目: Quantitative Analysis

**66.** Bertha is a risk manager who estimates the market risk of one of her firm’s portfolios. Because geometric returns are time-additive and arithmetic returns are cross-additive, she measures performance with both arithmetic and geometric returns. Her assumptions are gathered below:

- The portfolio value is \$100.0 million and she assumes there are 250 trading days per year.
- Using geometric returns, the average return +12.0% per annum with a standard deviation of 28.0%
- Using arithmetic returns, the average return +14.0% per annum with a standard deviation of 32.0%

She wants to measure the one-day 95.0% value at risk (VaR) under both approaches: assuming arithmetic returns are normal (aka, normal VaR) and assuming geometric returns are normal; aka, lognormal VaR. She assumes the returns are serially independent. Which of the following is **nearest** to the difference between the one-day 95.0% normal VaR and the one-day 95.0% lognormal VaR?

- A. The one-day 95% normal VaR is approximately \$37,700 less than the one-day 95% lognormal VaR
- B. The one-day 95% normal VaR is approximately \$380,100 less than the one-day 95% lognormal VaR
- C. The one-day 95% normal VaR is approximately \$29,400 greater than the one-day 95% lognormal VaR
- D. The one-day 95% normal VaR is approximately \$448,800 greater than the one-day 95% lognormal VaR

答案:D

解析:

**Option D is correct.** The one-day 95% normal VaR is approximately \$448,800 greater than the one-day 95% lognormal VaR. The one-day 95% normal VaR(%) is given by:

$$-14\%/250 + 32\% \times 1/\sqrt{250} \times 1.645 = 3.273246\%$$

The one-day 95% lognormal VaR(%) is given by:

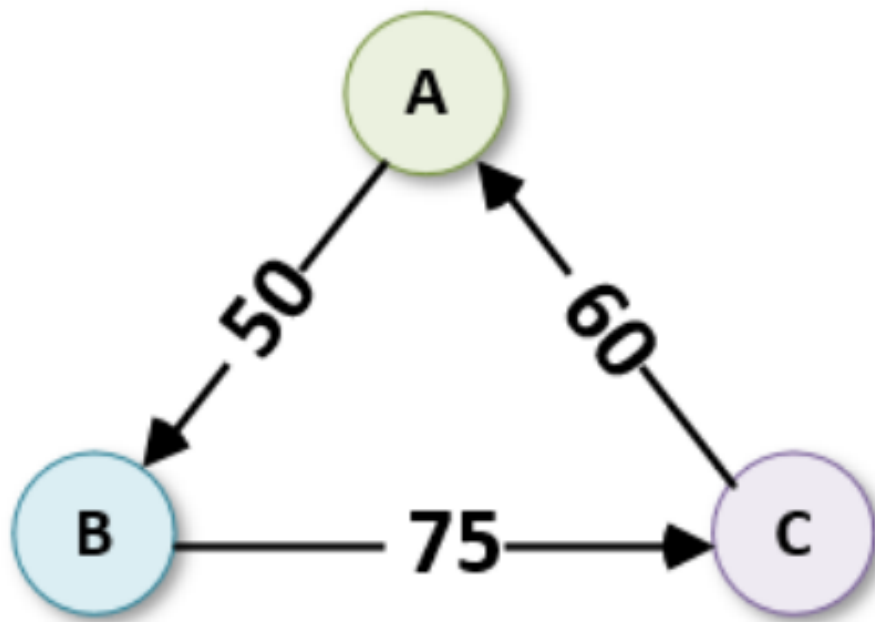
$$1 - \exp(12\%/250 - 28\% \times 1/\sqrt{250} \times 1.645) = 2.824436\%$$

The difference is  $3.273246\% - 2.824436\% = 0.448810\%$ , or about  $0.448810\% \times \$100.0 \text{ million} = \$448,810$ ; with an exact quantile, the difference is \$448,765.91.

考察: Quantitative Analysis: Value at Risk      重要程度: [■ ■ □]

科目: Quantitative Analysis

**67.** The graph below represents a trilateral scenario between three different counterparties (called A, B, and C) with position sizes in a certain fungible (aka, interchangeable) product. The values represent notional sizes and the direction represents exposures:



In this ring, the current total notional is 370. If trade compression can be achieved, which of the following is an efficient solution?

- A. a) A -> 25 -> B -> 60 -> C (new total notional = 110)
- B. b) A -> 30 -> B -> 45 -> C (new total notional = 150)
- C. c) B -> 35 -> C -> 30 -> A (new total notional = 65)
- D. d) B -> 25 -> C -> 10 -> A (new total notional = 70)

答案:D

解析:

**Option D is correct.** D. True: B -> 25 -> C -> 10 -> A (new total notional = 70)

The easiest approach is to identify the lowest exposure, in this case 50, and subtract it from all three. In this way,

- current A -> 50 -> B is eliminated;
- current B -> 75 -> C is reduced to B -> (75 – 50 = 25) -> C; and
- current C -> 60 -> A is reduced to C -> (60 – 50 = 10) -> A

考察: Finance    重要程度: [■ ■ □]

科目: Quantitative Analysis

**68.** Sally is a risk analyst at an international bank who is leading a project to conduct a causal analysis of the bank’ s operational risk. The project will coordinate with, and inform modifications to, the bank’ s loss database. One of the project’ s goals is to analyze the organization’ s controls, control layers, and key risk indicators (KRIs), and key control indicators (KCIIs).

She prefers to use a factor model, and her candidates include the following approaches:

- *Fault tree analysis (FTA)*: this deductive method will build a series of and/or conditions that can happen simultaneously or alternatively
- *Root-cause analysis with bowtie tool*: the visual tool will employ the 5-why analysis to probe deeply into the root causes of events
- *FAIR methodology*: this structured framework will generate scenarios, where each scenario will assume an asset at risk, a threat community, a threat type, and an effect

- *Swiss cheese model*; aka cumulative act effects: by anticipating active failures and identifying latent conditions, this model will promote effective control layering and identifying active failures

Each of these four approaches has advantages and disadvantages. For ANY of these approaches, which of the following paired statements (i.e., a single advantage paired with a single disadvantage) is a TRUE statement?

- A. Fault tree analysis (FTA) is simple and non-reliant on domain expertise; however, it is unrealistic because it can only assume independent events
- B. The Bowtie tool is proactive because it assesses the likelihood and/or impact of risks before they materialize by examining controls and indirect causes; however, it natively lacks a quantitative framework such that prioritization is harder
- C. The FAIR model is inclusive because it emphasizes subjective identification (aka, findability) of factors; however, it is not a probabilistic approach and may be insufficient when distributions are required
- D. The Swiss cheese model is elegant because clusters and their centroids (aka, holes) are quantitatively retrieved via k-means clustering; however, it struggles when incorporating human factors into a holistic risk perspective

答案:B

解析:

**B 选项正确。** True: The Bowtie tool is proactive because it assesses the likelihood and/or impact of risks before they materialize by examining controls and indirect causes; however, it natively lacks a quantitative framework such that prioritization is harder In regard to (A), (C), and (D), each is false.

考察: Operational Risk      重要程度: [■ ■ □]

科目: Financial Markets and Products

**69.** According to Rose and Hudgins, the factors affecting the choice of non-deposit funding sources include the following: the relative costs of each funding source; the risk of each funding source, including the dependability and volatility; the length of time for which funds will be needed; the size of the financial institution needing the funds; and regulations on the use of other sources of funds. In regard to the various sources of non-deposit funding, which of the following statements is TRUE?

- A. The Fed Funds market is the least popular because its effective interest rate is higher than the alternatives
- B. Repurchase agreements are more popular than Fed Funds because they are simpler, however, they incur greater credit risk
- C. The advantages of the commercial paper (CP) market are the lack of credit risk and consistently high (i.e., low volatility) credit availability, however, volume tends to be low and relative cost is high
- D. Advances from Federal Home Loan Banks (aka, FHLBanks) are highly popular among thrifts and small banks as a stable source of flexible (with respect to maturity) funding at below-market interest rates

答案:D

解析:

**D 选项正确。** Advances from Federal Home Loan Banks (aka, FHLBanks) are highly popular among thrifts and small banks as a stable source of flexible (with respect to maturity) funding at below-market interest rates.

From the Study Notes: “Advanced from Federal Home Loan Banks: In recent years the Federal Home Loan Bank (FHLB) System has moved into lending money to home mortgage lenders. Comprising 12 regional banks, the FHLB was created in 1932 as a source of stable funding for depository institutions facing potential runs by depositors. According to many scholars, the involvement of FHLB in the “sources of funds” arena has not only increased the amount of available funds for financial institutions that use FHLB as a source (with mortgages as collateral), but FHLB has also increased the liquidity of home loans and provided an incentive for more financial institutions to provide credit to the housing market.

The presence of FHLB has quickly expanded in recent years, with over 6,000 commercial banks, over 1,300 thrift institutions, over 700 credit unions, and over 80 insurance companies now eligible to borrow from the FHLB. The draw to FHLB funds stems from it being a stable source of finance and offering below-market interest rates. Financial institutions can borrow from the FHLB for a flexible time period, ranging from overnight to over 20 years, with fixed or variable interest rates. Part of the reason the FHLB can offer below-market interest rates is because its federal charter allows it to borrow money on the cheap and

pass those savings on to member banks, who are stockholders and receive dividends. The Federal Home Loan Banks System also has a favorable position in the event of failure. Legally, the FHLB is first in line, even in front of the FDIC, to recover funds if a failure occurs.”

In regard to (A), (B) and (C), each is FALSE. Instead:

- The Fed Funds market is the most popular and its effective interest rate tends to be LOWER than the alternatives, although it is VOLATILE (is its downside)
- Repurchase agreements are less popular and more complex than Fed Funds; also, they incur less credit risk because they are secured by collateral (they are often viewed as “collateralized Fed funds transactions” ).
- Commercial paper (CP) is high in volume, moderate in cost (advantages) but volatility in available capacity and subject to credit risk (disadvantages)

考察: Finance: Non-deposit Funding Sources      重要程度: [■ ■ □]

科目: Quantitative Analysis

**70.** We can add a momentum factor to the Fama-French so that it becomes a four-factor model. This momentum factor is denoted by WML (i.e., past winners minus past losers) or UMD (i.e., stocks that have gone up minus stocks that have gone down). At least with respect to the historical window analyzed, which is the long period from January 1965 to December 2011, which of the following statements is TRUE about the momentum factor?

- A. Momentum is a negative feedback strategy which is inherently stabilizing
- B. The momentum factor is observed in equities but is NOT observed in bonds, commodities, and real estate
- C. Momentum investing by definition is an anti-value strategy; correlations between HML and WML are strongly negative
- D. The cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on either size or value

答案:D

解析:

**Option D is correct.** The cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on either size or value.

In regard to (A), (B) and (C), each is false.

考察: Factor Investing      重要程度: [■ ■ □]

科目: Quantitative Analysis

**71.** Peter has collected the historical sample of daily profit and loss (P/L) observations for his firm’s new equity portfolio but his dataset only contains 250 days. Ideally, he would prefer 1,000 observations because the board asked for a presentation that displays value at risk (VaR) at confidence levels of (99.0%, 99.1%, ..., 99.8%, and 99.9%). The board also expressed a preference for a non-parametric approach; apparently, some directors are concerned that parametric approaches might be too difficult to communicate and understand. In addition to granular VaR confidence levels (e.g., 99.1%), if possible, the board would like to see confidence INTERVALS around the VaR estimates in order to gauge their precision; e.g., a 9X% confidence interval around the 99.1% VaR.

Therefore, Peter considers the following four approaches: basic historical simulation (BHS), bootstrapped historical simulation (BOOT), historical simulation using non-parametric density estimation (NPDE), and age-weighted historical simulation (AWHS). Given the realities of his relatively small, empirical dataset, which of the following statements is TRUE?

- A. Peter can easily eliminate the BHS approach because basic historical simulation cannot even retrieve a 99.10% (or 99.20% etc.) VaR quantile due to the fact that 1/250 is greater than 0.10%
- B. Peter has a dilemma (i.e., none of the four approaches is feasible) as the construction of confidence levels around VaR estimates will require him to employ a parametric (or semi-parametric at least) approach



- C. Peter has a dilemma (i.e., no approach is good here) simply because his sample size is too small to meet the board's preferences under any of the four approaches
- D. Non-parametric density estimation (NPDE) can retrieve VaRs at granular confidence levels (e.g., 99.10%) as if the distribution is a continuous probability density function (pdf)

答案:D

解析:

**D 选项正确。** Non-parametric density estimation (NPDE) can retrieve VaRs at granular confidence levels (e.g., 99.10%) as if the distribution is a continuous probability density function (pdf)

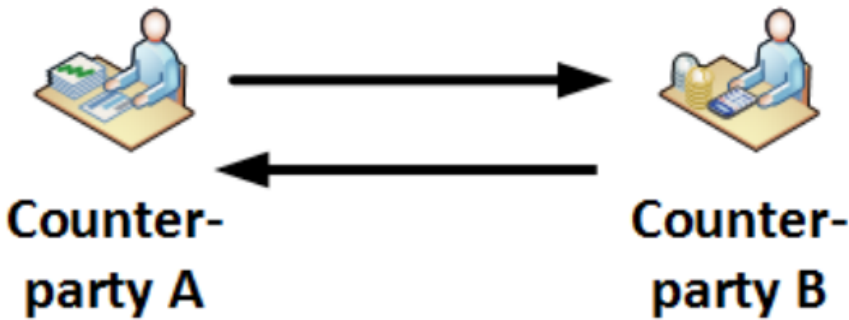
In regard to (A), (B), and (C), each is FALSE. Instead, the following are TRUE statements:

- Basic historical simulation definitely does return a 99.1% VaR here it will (probably) be the same as the 99.0% VaR
- Due to the convenient power of order statistics, confidence intervals can be generated under all four approaches; or really, confidence intervals can be built for any parametric or non-parametric VaR or ES estimate!
- The small sample size might favor the bootstrap or the non-parametric density estimation

考察: Market Risk Measurement and Management      重要程度: [■■■]

科目: Quantitative Analysis

**72.** Two counterparties (A and B, illustrated below) have between them a set of OTC derivative transactions.



Associated with the positions, these counterparties also entered into a bilateral collateral agreement prior to initiation of the underlying transactions. Each of the following is true EXCEPT which is false?

- A. If they followed best practice, collateral posting should be continuous for both of them
- B. As a counterparty risk mitigant, collateral posting is generally superior to a reset feature
- C. Variation margin is a type of collateral that aims to track the change in valuation (aka, change in MTM) of the position over time
- D. As a bilateral agreement, in the case of a positive MTM, the party will call for collateral; and in the case of negative MTM, the party will be required to themselves post collateral

答案:A

解析:

**A 选项正确。** False. Continuous posting of collateral is operationally too expensive In regard to (B), (C) and (D), each is TRUE.

考察: Counterparty Credit Risk      重要程度: [■■■]

科目: Quantitative Analysis

**73.** MetroScape Bank prides itself on its modern banking solutions and customer-centric approach. During the last quarter, however, a series of events occurred that suggest the bank's controls should (or at least can) be improved. Specifically, each of the four major types of human error was observed:

- I. Jane, a seasoned cashier at MetroScape, had been handling cash transactions for years. One busy Friday afternoon, while multitasking between customer queries and cash handling, she inadvertently gave out \$500 more to a customer than he had withdrawn. Her muscle memory, developed from years of handling similar transactions, failed her in that moment of distraction.
- II. MetroScape recently redesigned its compensation and incentive programs. The incentive program rewards cross-selling, especially when a valuable customer adds an additional account and/or utilizes a new program. Due to this incentive plan, some branches aggressively cross-sold products to existing customers without sufficient regard for product suitability. Several customers called headquarters to complain about the sales tactics.
- III. A new investment product was introduced at MetroScape, and the bank's financial advisors were tasked with guiding interested clients. Tom, one of the advisors, faced a unique situation where a client wanted to merge this new product with another existing one. Unsure and without prior experience or training in this scenario, Tom made an educated guess. Unfortunately, his decision led to significant tax implications for the client.
- IV. Lucy, a senior executive at MetroScape, was aware of the bank's policy against insider trading. However, tempted by the prospect of significant personal gain, she used confidential information about an upcoming merger to buy shares in a company that MetroScape was about to acquire. Her actions, driven by personal greed, were a clear violation of both the bank's policy and federal regulations.

In regard to these events and their associated control implications, each of the following statements is true EXCEPT which is false?

- A. The fourth (IV.) is a violation which is best mitigated by supervisory controls
- B. The third (III.) is a type of knowledge-based (KB) mistake which can be mitigated by directive controls
- C. The second (II.) is a quality-based (QB) mistake where the best mitigation is typically lean Six Sigma response
- D. The first (I.) is a skill-based (SB) slip, and root-cause analysis may imply process re-engineering, environmental factors, and/or workflow changes

答案:C

解析:

**C 选项正确。** False. This is a knowledge-based (KB) mistake. According to the reading (GARP), the relevant response(s) to knowledge-based mistakes are directive controls.

In regard to (A), (B), and (D), each is TRUE.

考察: Operational Risk: Human Error      重要程度: ■■■□

科目: Quantitative Analysis

**74.** What is the primary (valid) reason that banks need proper liquidity transfer pricing (LTP)?

- A. Because banks engage in maturity transformation
- B. Because there is a lack of liquidity transformation products in the marketplace
- C. Because the true economic cost of liquidity risk is approximately zero
- D. Because markets are volatile; e.g., volatility 1-year LIBOR/Swap spread

答案:A

解析:

**A 选项正确。** True: Because banks engage in maturity transformation (maturity transformation is the recurring causal theme in the reading).

*As Grant explains (in 3.1), Why banks need LTP:* In their daily operations, banks make money by funding long-term loans (assets) with short-term deposits (liabilities), a process that is commonly referred to as maturity transformation. As pointed out by the BCBS (September 2008), this “makes banks inherently vulnerable to liquidity risk, both of an institution-specific

nature and that which affects markets as a whole” (p 3). But provided banks use LTP to account for the costs, benefits and risks of liquidity in product pricing, new product approval processes and profit and performance assessments, they should not be discouraged from engaging in maturity transformation. Banks with poor LTP practices are more likely to accrue larger amounts of long-term illiquid assets, contingent commitments and shorter-dated volatile liabilities, substantially increasing their vulnerability to funding shortfalls.” (Joel Grant, 2011. “Liquidity Transfer Pricing: A Guide to Better Practice,” Occasional Paper, Financial Stability Board, Bank for International Settlements.)

In regard to (B), (C) and (D) each is FALSE.

考察: Liquidity Risk Management      重要程度: ☒☒☐

科目: Quantitative Analysis

**75.** In the paper “Finding Bernie Madoff” (by Dimmock and Gerken), the authors produce several findings as they attempt to test the predictability of investment fraud. In regard to their findings specifically, each of the following is TRUE except which is not?

- A. Disclosed soft dollars do not significantly predict fraud
- B. Direct Theft and Self Dealing are the most common types of fraud
- C. Current disclosures required by the SEC have significant power for predicting future fraud
- D. Investors (on average) are compensated for fraud risk through superior performance and/or higher fees

答案:D

解析:

**D 选项正确。** FALSE. Instead, the authors find that investors are NOT compensated for fraud risk, either through superior performance and/or higher fees

In regard to (A), (B), and (C), each is TRUE.

考察: Operational Due Diligence      重要程度: ☒☒☐

科目: Quantitative Analysis

**76.** Ralph employed the peaks-over-threshold (POT) approach under extreme value theory (EVT) to the estimation of value at risk (VaR) for his firm’ s portfolio. He has four years of data,  $n = 1,000$  days. He would prefer to apply a maximum likelihood estimation method (aka, MLE) to parameter estimation, but he thinks he needs to choose the threshold first. When he selects a threshold of  $u = 4.0\%$ , he observes  $N(u) = 28$  exceedances such that  $N(u)/n = 2.80\%$ . Fitting the data generates these parameters: scale,  $\beta = 0.80\%$ ; and tail index,  $\xi = 0.310$ . Based on these assumptions, the 99.90% confident VaR is 8.670%. He subsequently decides to increase the threshold from 4.0% to 5.0% while retaining the tail index at 0.310 and the scale parameter at 0.80%. The number of exceptions declines to  $N(u) = 17$  such that  $N(u)/n = 17/1,000 = 1.70\%$ . His revised 99.90% VaR is similar at 8.630%. In regard to choosing a threshold, which of the following statements is TRUE?

- A. The higher threshold offers the benefit of lower bias but the drawback of higher variance
- B. The higher threshold offers the benefit of lower variance but the drawback of higher bias
- C. Because the desired confidence is 99.90% and  $0.10\% = 1/1,000$ , the (optimal) threshold should be set equal to the dataset’ s second-worst return
- D. He can avoid a subjective choice of the threshold by applying an analytical maximum log-likelihood estimation (MLE) function to all three parameters simultaneously

答案:A

解析:

**A 选项正确。** We confront a bias-variance tradeoff in either EVT approach. In the blocks approach, the tradeoff is a reflection of the size (or number) of blocks. In the POT approach, it is a reflection of the arbitrary threshold choice, which is analogous to the block size choice. If the threshold is too low, then our model will not comport with EVT’ s asymptotic properties and the

model is likely to be inaccurate (aka, biased). A higher threshold implies lower bias (aka, greater accuracy) but this necessarily fits to fewer data points such that we can expect higher variance (aka, less precision).

Dowd<sup>(†)</sup> references the bias-variance tradeoff in the context of semi-parametric estimation methods (e.g., Hill estimator): “... procedure based on the fact that the choice of  $k$  implies a trade-off between bias and variance. If we increase  $k$ , we get more data and so move to the centre of the distribution. This increases the precision of our estimator (and therefore reduces its variance), but also increases the bias of the tail estimator by placing relatively more weight on observations closer to the centre of our distribution. Alternatively, if we decrease  $k$  and move further out along the tail, we decrease bias but have fewer data to work with and get a higher variance.”

考察: Quantitative Analysis: Extreme Value Theory      重要程度: [■■■□]

科目: Quantitative Analysis

**77.** Consider two counterparties who are about to enter into bilateral framework under an ISDA Master Agreement. They are currently negotiating the credit support annex (CSA) which will govern the terms of collateral posting. Each of the following is true EXCEPT which is inaccurate?

- A. If the goal is to minimize funding costs, both parties prefer to avoid rehypothecation rights
- B. With respect to VARIATION margin, both parties probably prefer rehypothecation rights; and probably will NOT require segregation
- C. With respect to INITIAL margin, both parties probably are likely to insist on non-rehypothecation; and probably will require segregation
- D. Whoever is the collateral poster (aka, collateral giver) prefers the method of Security Interest rather than the method of Title Transfer (which is preferred by the collateral receiver)

答案:A

解析:

**A 选项正确。** A. FALSE. Rehypothecation minimizes funding cost. The trade-off is that rehypothecation minimizes funding cost but increases counterparty risk.

In regard to (B), (C) and (D), each is TRUE.

考察: Counterparty Risk      重要程度: [■■■□]

科目: Quantitative Analysis

**78.** Reporting on operational risks presents a unique set of challenges. As GARP explains, “Unlike other risk factors, assessing the volatility of operational risk is especially challenging given this is a relatively new discipline. Credit and market risk metrics often have over a hundred years of historical trend information. Operational risk measurement is relatively new, requiring more attention to the qualitative and quantitative tracking measures used to evaluate changes in risk profiles.”<sup>(†)</sup>

Let’ s assume the organization’ s central operational risk function (ORF) has primary responsibility for operational risk reporting. In regard to the challenges to reporting operational risks, each of the following statements is true EXCEPT which is false?

- A. The ORF should focus scarce resources on preventing and addressing major incidents rather than getting distracted by daily volatility
- B. When the purpose is benchmarking across business units, losses should be reported in dollar terms (i.e., rather than percentage or basis points) because low percentages could mask high dollar losses
- C. The ORF should regularly review small, frequent losses to spot potential control breaches or process flaws; if these losses are consistent and random, consider including their average cost in the service price
- D. The ORF can address the challenge of compiling qualitative risk data through either (i) conversion and summation, (ii) categorization, and (iii) worst-case reporting.

答案:B

解析:

**B 选项正确。** False. Instead, percentages are preferred to dollars.

Says GARP, “Reporting operational losses as a percentage of gross income, of total cost (for a cost center), total budget (for a project), or in basis points of capital facilitates the comparability of similar entities across business units. Reporting in percentages instead of absolute values also makes the reporting scalable and applicable to entities of all sizes. Ideally, operational risk managers will also better understand the cost-benefit assessment and trade-offs related to implementing the appropriate controls and operating restrictions.”(†)

In regard to (A), (C) and (D), each is TRUE.

- The ORF should focus scarce resources on preventing and addressing major incidents rather than getting distracted by daily volatility. Says GARP, “A proper allocation of risk management resources is to focus on the prevention and remediation of large incidents and not to be dispersed and distracted by the containment of daily volatility”(†)
- The ORF should regularly review small, frequent losses to spot potential control breaches or process flaws; if these losses are consistent and random, consider including their average cost in the service price. Says GARP, “Small and frequent losses should be identified and analyzed regularly to detect any pattern that could signal a breach in control or a structural flaw in a process that would warrant an action plan. If these losses occur randomly, without any specific structure, and are structurally limited, stable, and repetitive, their average cost can be passed through to the customers as part of the cost of services.”(†)
- The ORF can address the challenge of compiling qualitative risk data through either (i) conversion and summation, (ii) categorization, and (iii) worst-case reporting.

(†) GARP Chapter 6, Risk Reporting, 2023 FRM Part 2, Topic 7.

考察: Operational Risk Reporting      重要程度: [■ ■ □]

科目: Quantitative Analysis

**79.** According to Patrick McGuire, which of the following best summarizes the proximate cause of the US dollar shortage during the global financial crisis (GFC)?

- A. Depositors initiated a classic run on the banks and converted their domestic deposits to foreign currency
- B. The effective maturity of banks’ dollar funding lengthened while their dollar assets shortened, and this decreased their maturity mismatch.
- C. The effective maturity of banks’ dollar funding shortened while their dollar assets lengthened, and this increased their maturity mismatch.
- D. Banks shifted their portfolios to non-bank assets that included banks’ retail and corporate lending, lending to hedge funds, and holdings of securities ranging from US Treasury and agency securities to structured finance products

答案:C

解析:

**Option C is correct.** The effective maturity of banks’ dollar funding shortened while their dollar assets lengthened, and this increased their maturity mismatch.

McGuire (emphasis ours): “Events during the crisis led to severe disruptions in banks’ sources of short-term funding. Interbank markets seized up, and dislocations in FX swap markets made it even more expensive to obtain US dollars via swaps. Banks’ funding pressures were compounded by instability in non-bank sources of funds as well, notably dollar money market funds and dollar-holding central banks. The market stress meant that the effective maturity of banks’ US dollar funding shortened just as that of their US dollar assets lengthened, since many assets became difficult to sell in illiquid markets. This endogenous rise in maturity mismatch, difficult to hedge ex ante, generated the global US dollar shortage. Our estimate of the size of banks’ US dollar funding gaps at the onset of the crisis shed light on why the international policy response was necessary, and why it took the form of a global network of central bank swap lines.” (Patrick McGuire, Gotz von Peter, 2009. “The US Dollar Shortage in Global Banking and the International Policy Response,” BIS Working Papers, Bank for International Settlements.)

考察: Finance: Global Banking      重要程度: [■ ■ □]

科目: Quantitative Analysis

80. Janice is preparing to conduct factor regressions in order to discover the factor loadings in her firm’s equity portfolio. She will begin by regressing three explanatory (aka, independent) variables: the market factor (MKT) plus two additional factors. Each of the following pairs of factors are good candidates for these additional factors EXCEPT which of the following pair is NOT a good candidate?

- A. Quality and momentum (style factors)
- B. Strategy and reputation (style factors)
- C. Economic growth and inflation (macro factors)
- D. Real interest rates and liquidity (macro factors)

答案:B

解析:

B 选项正确。 False. In reality, it is difficult to invest directly in reputation, and it seems almost impossible to imagine how to operationalize strategy as an investable factor.

According to Andrew Ang (see <https://www.blackrock.com/us/individual/investment-ideas/what-is-factor-investing>) there are two primary types of factors: macroeconomic factors and style factors. Macroeconomic factors include economic growth, credit (default risk), real interest rates, emerging markets (political and sovereign risk), inflation, and liquidity. Style factors include value, quality, minimum volatility, size, momentum, and carry.

In this regard, each of (A), (C) and (D) is TRUE:

- Quality and momentum are genuine style factors
- Economic growth and inflation are genuine macro factors

考察: Factor Investing      重要程度: [■ ■ □]

科目: Quantitative Analysis